Fractal Geometry

In mathematics and in nature

Lindelöf Math Society

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underlying meta-text:

What is Mathematics?

Is mathematics a set of logically interconnected statements (which are miraculously suited to describe the world)?

or is mathematics a LANGUAGE and an ACTIVITY through which we interact with and describe the world around us?

underlying meta-text:

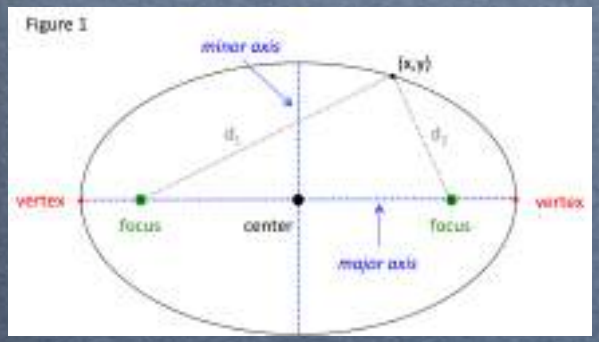
How to Learn Mathematics?

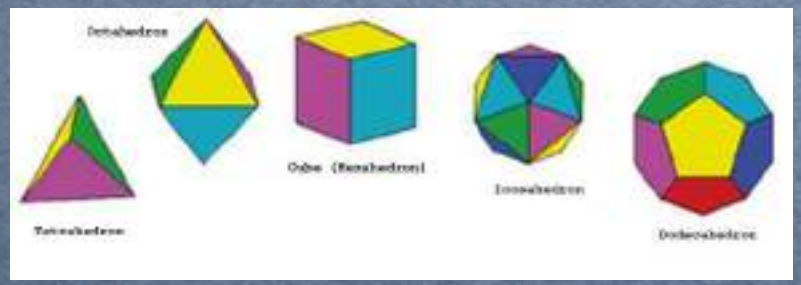
as a LANGUAGE and an ACTIVITY
the only way to learn mathematics is by
DOING and PRACTICING!

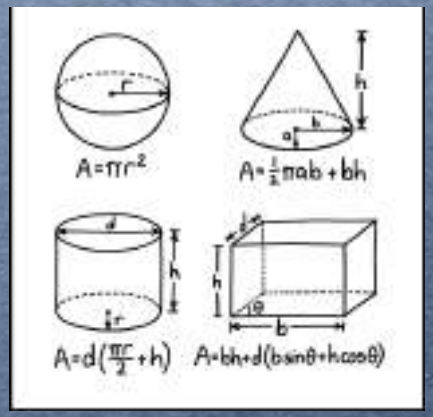
Mathematics is like COOKING: you can read hundreds of cookery books but will never learn to cook, let alone INVENT NEW RECIPES, unless you actually COOK!

Geometry: the study of shapes

What you think about when you think of geometry







vs. what is out there:

Benoit Mandelbrot
(1924-2010)

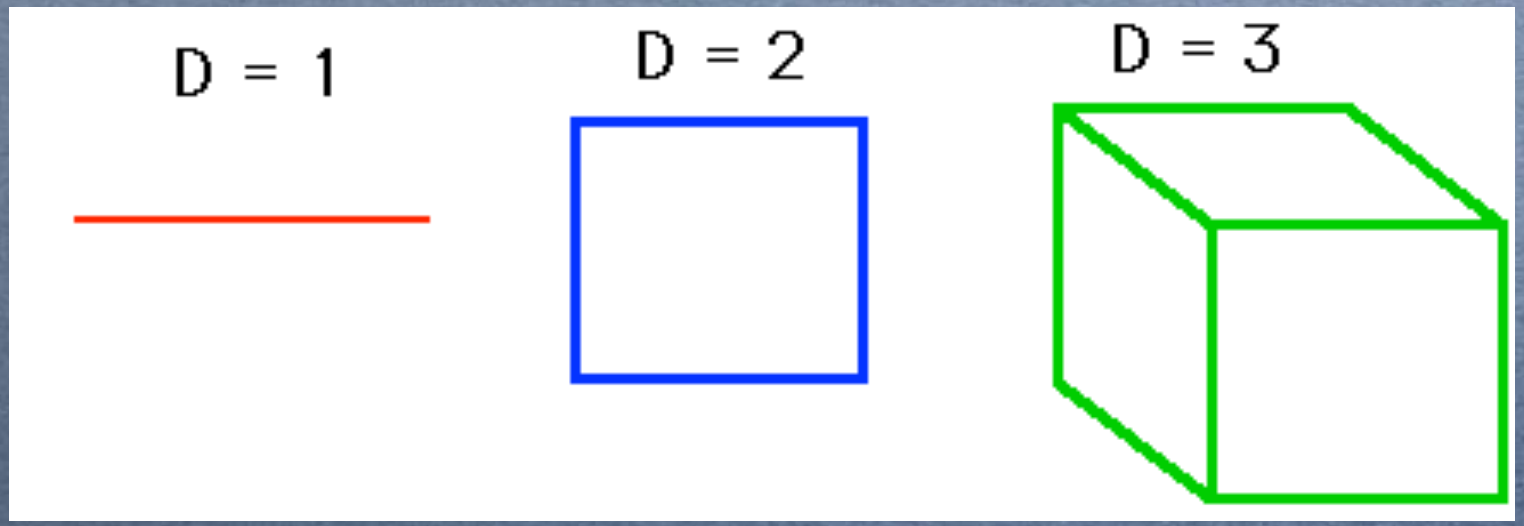
The fractal geometry of nature (1982)

"I claim that many patterns of Nature are so irregular and fragmented, that, compared with [standard geometry], Nature exhibits not simply a higher degree but an altogether different level of complexity ... "

As quoted in a review of *The Fractal Geometry of Nature* by J. W. Cannon in *The American Mathematical Monthly*, Vol. 91, No. 9 (November 1984), p. 594

A core idea is DIMENSION

What is DIMENSION?



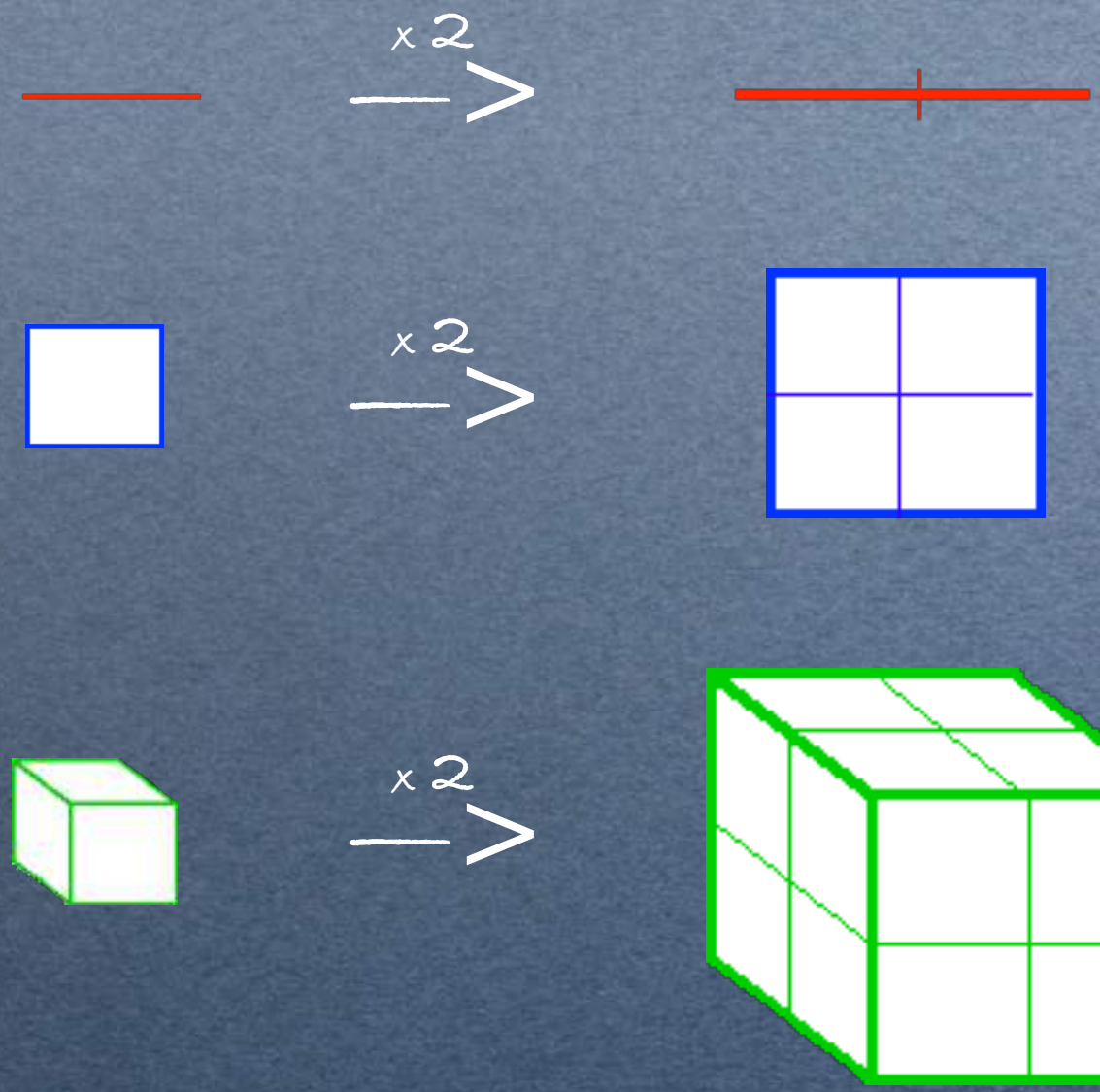
Dimensions greater than 3?

Non-integer dimensions?

What is DIMENSION?

Several ways to generalize the concept of dimension to make sense of the non-integer (fractional) dimensions.

- Self-Similarity Dimension (only in certain situations)
- Hausdorff Dimension

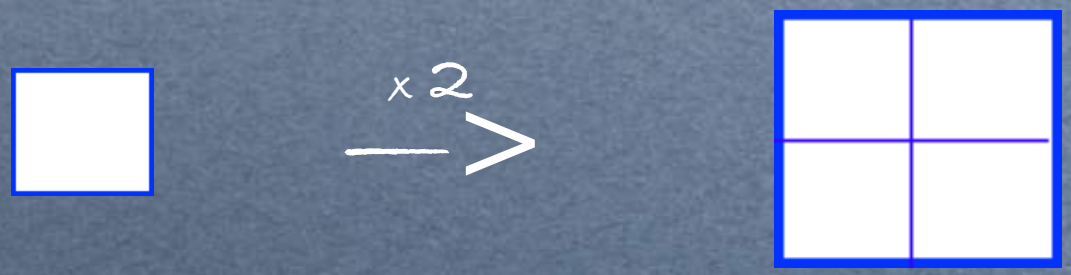


2 copies
of the original

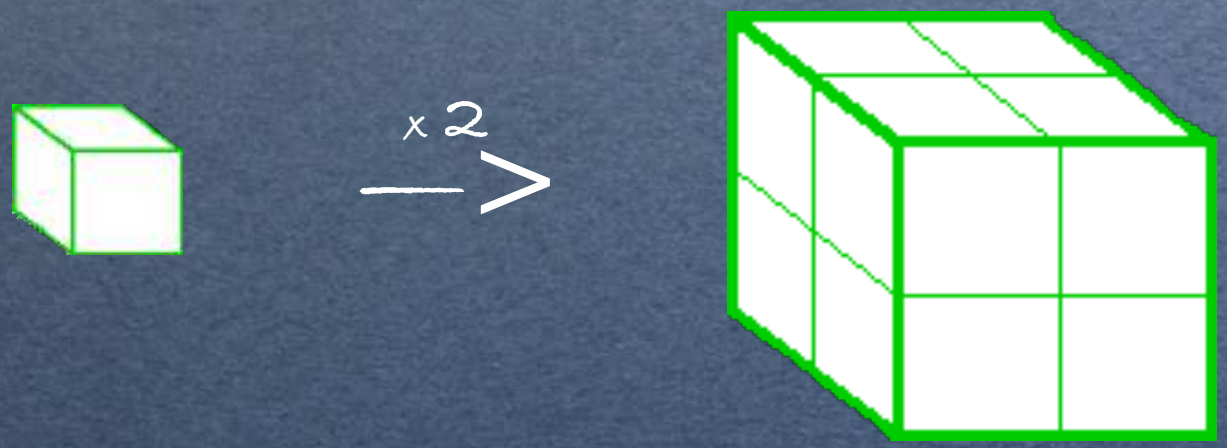
4 copies
of the original

8 copies
of the original

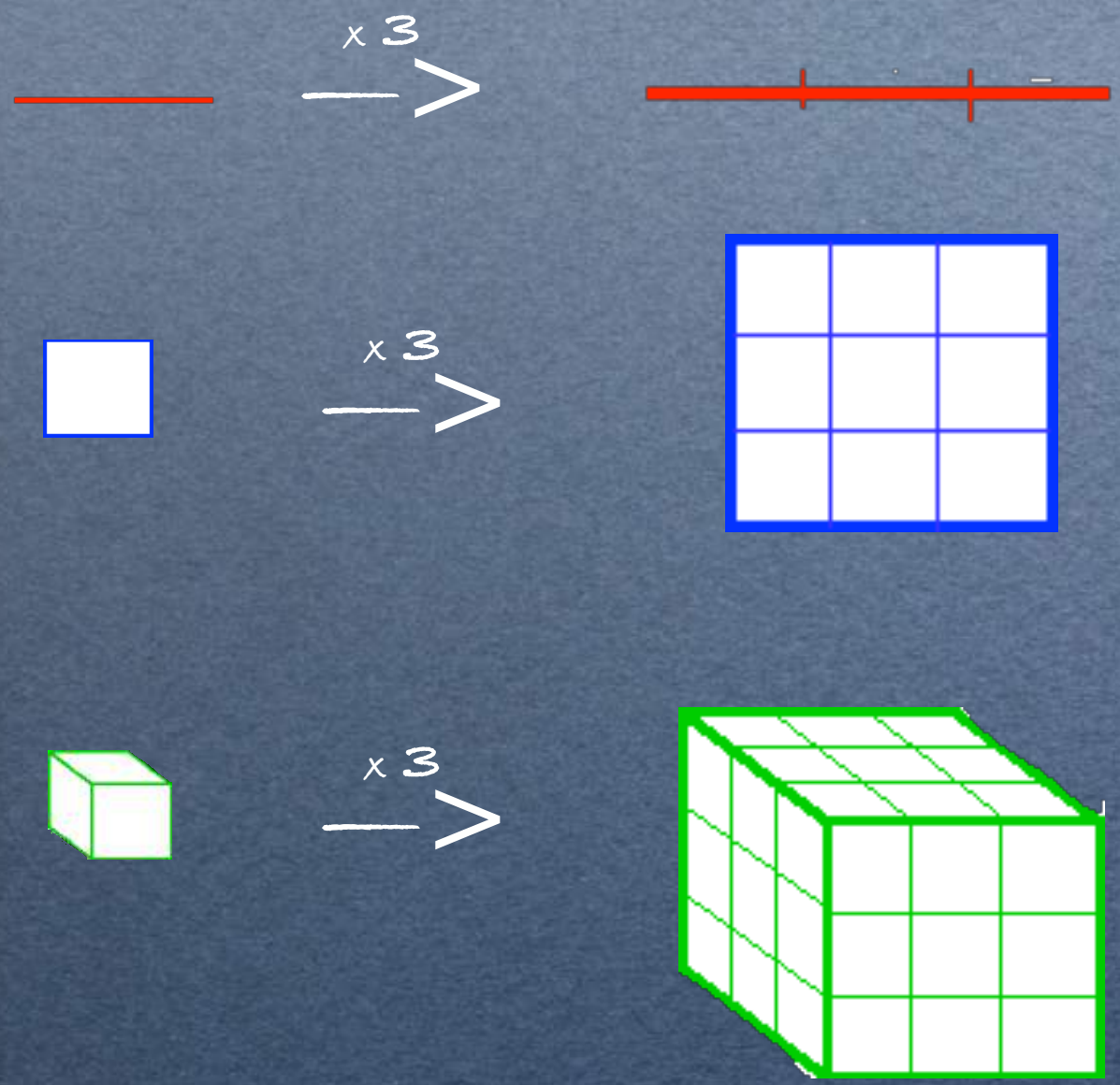
$$2 = 2^1$$



$$4 = 2^2$$



$$8 = 2^3$$

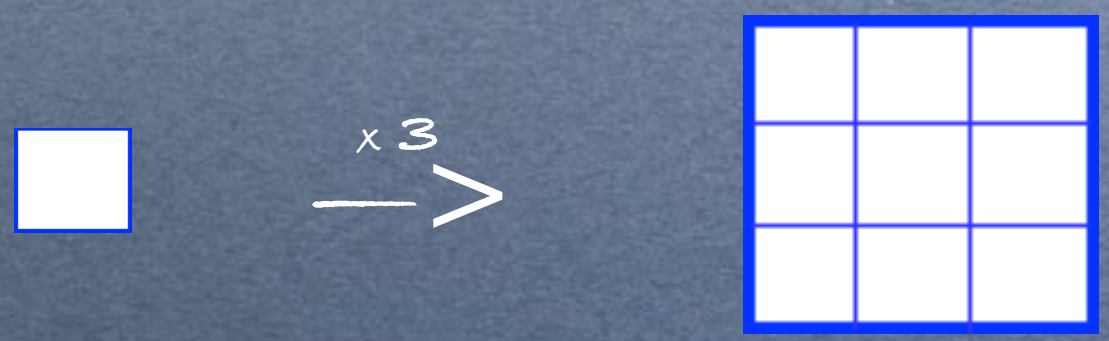


3 copies
of the original

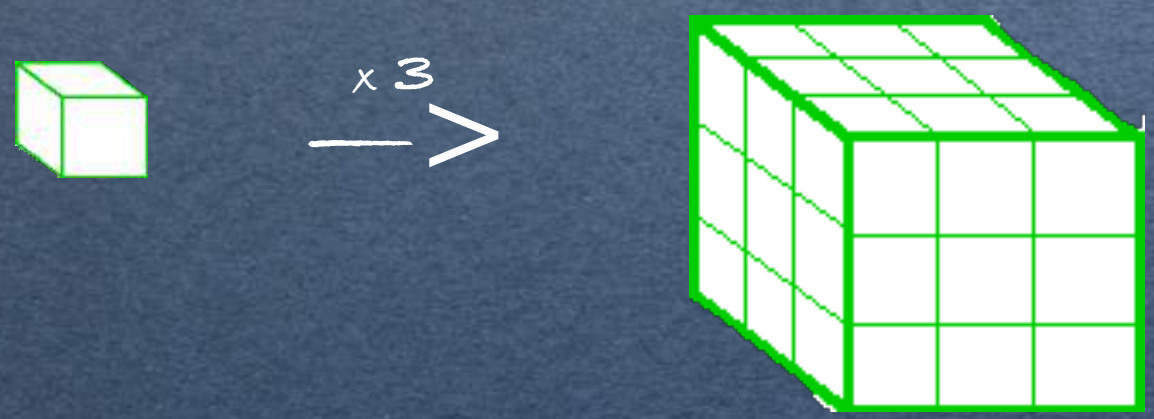
9 copies
of the original

27 copies
of the original

$3 = 3^1$



$9 = 3^2$



$27 = 3^3$

If dimension is d, a scaling by a factor k contains

$$N = k^d$$

copies of the original

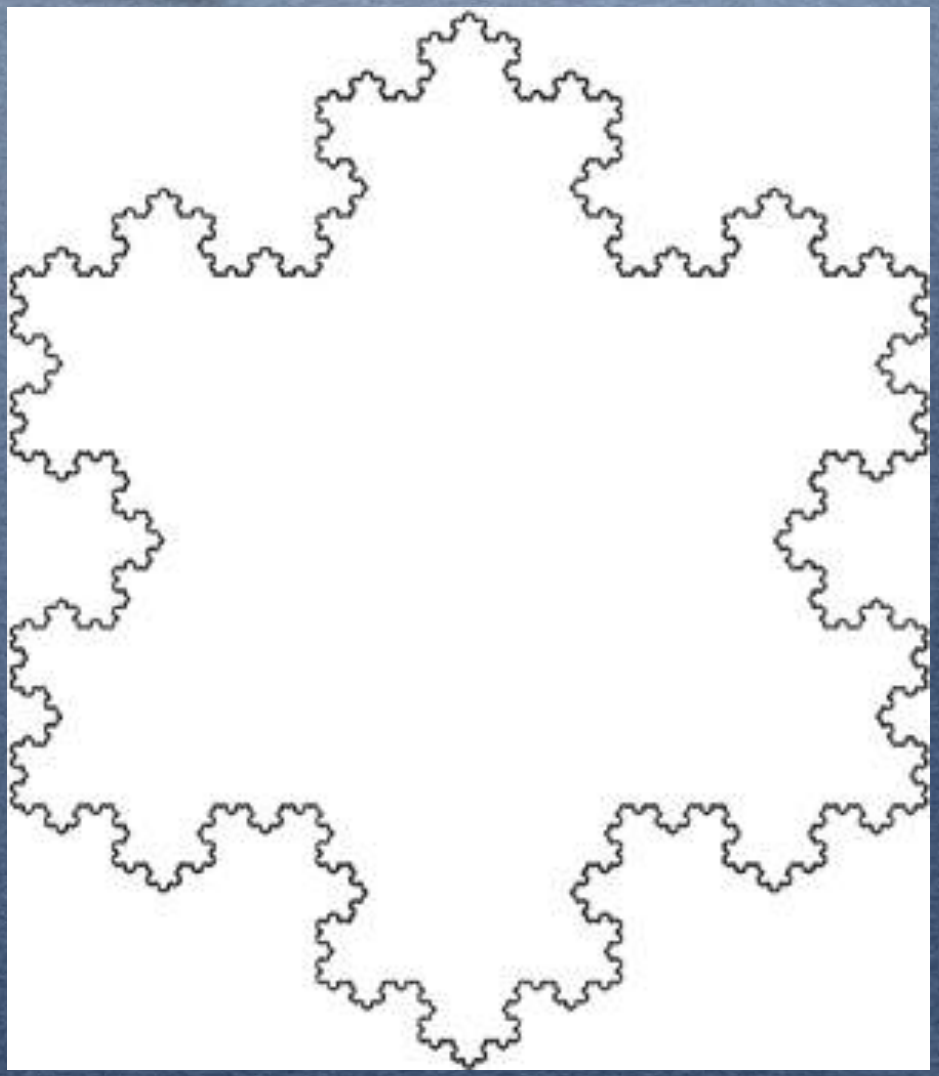
Self-Similarity Dimension

$$d = \log N / \log k$$

If a shape scaled by a factor k contains N copies of the original then the dimension is d

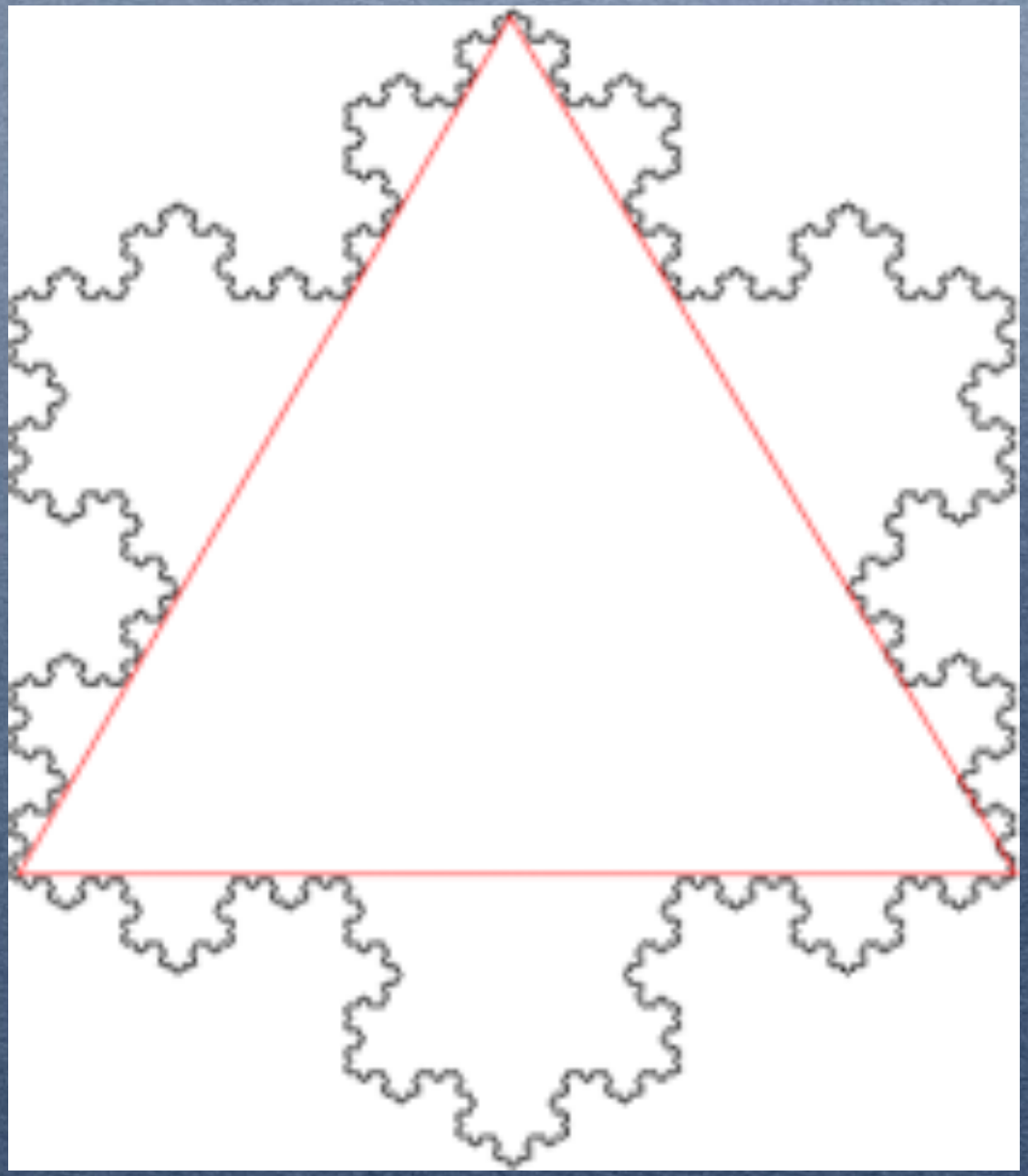
Example: The Snowflake

A "curve" with infinite detail



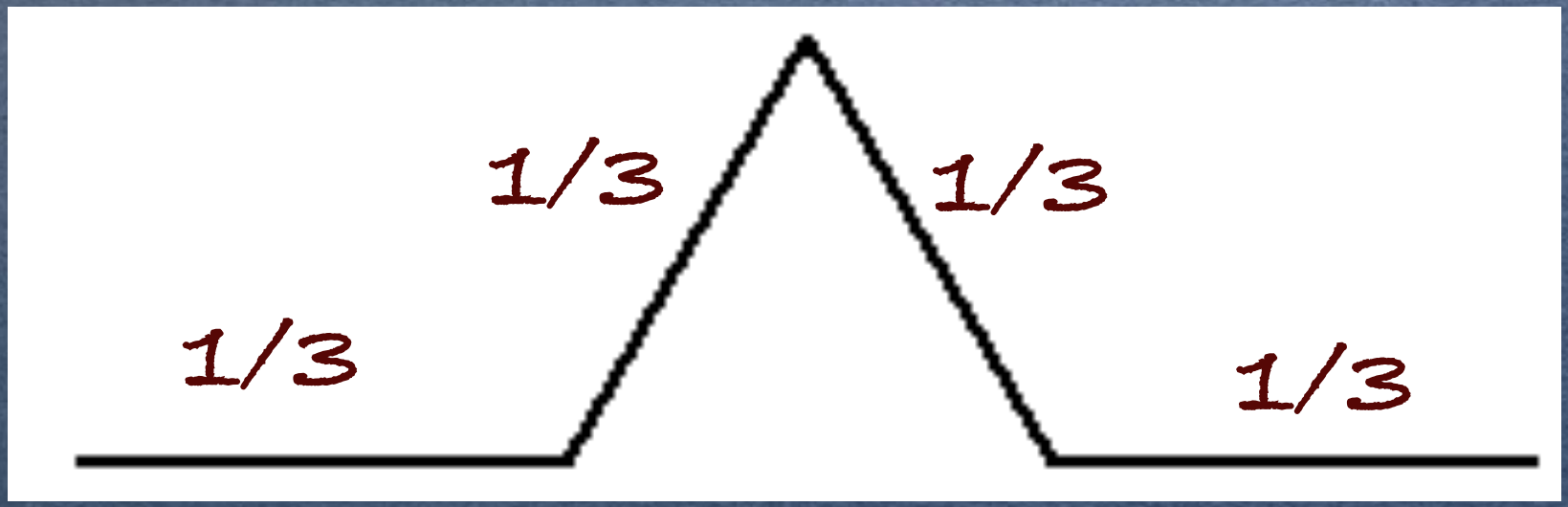
Sur une courbe continue sans tangente, obtenue par une construction géométrique élémentaire
Helge von Koch, 1904

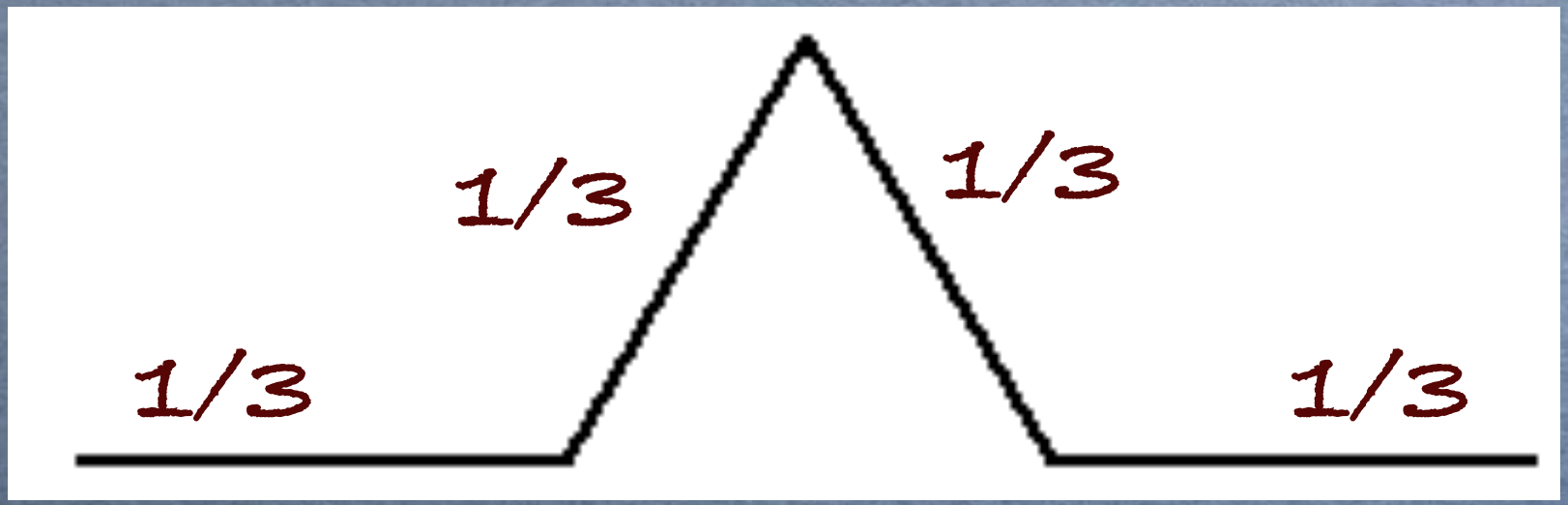
Example: The Snowflake



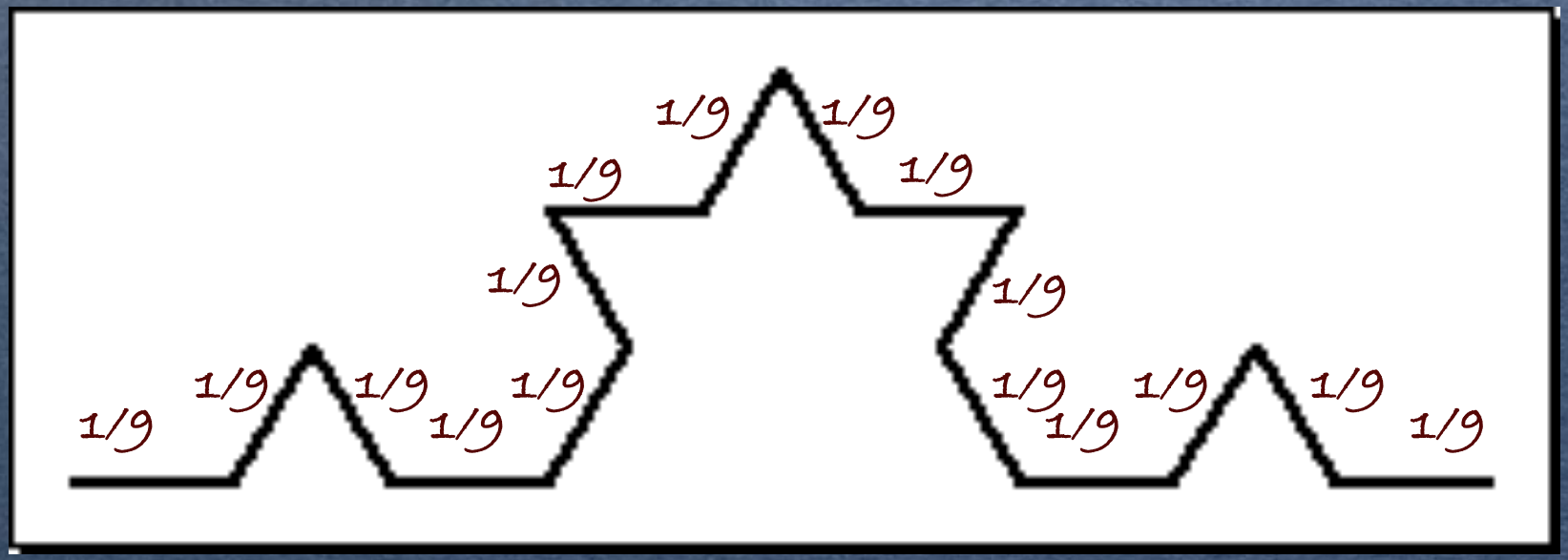
Start with one side of the triangle

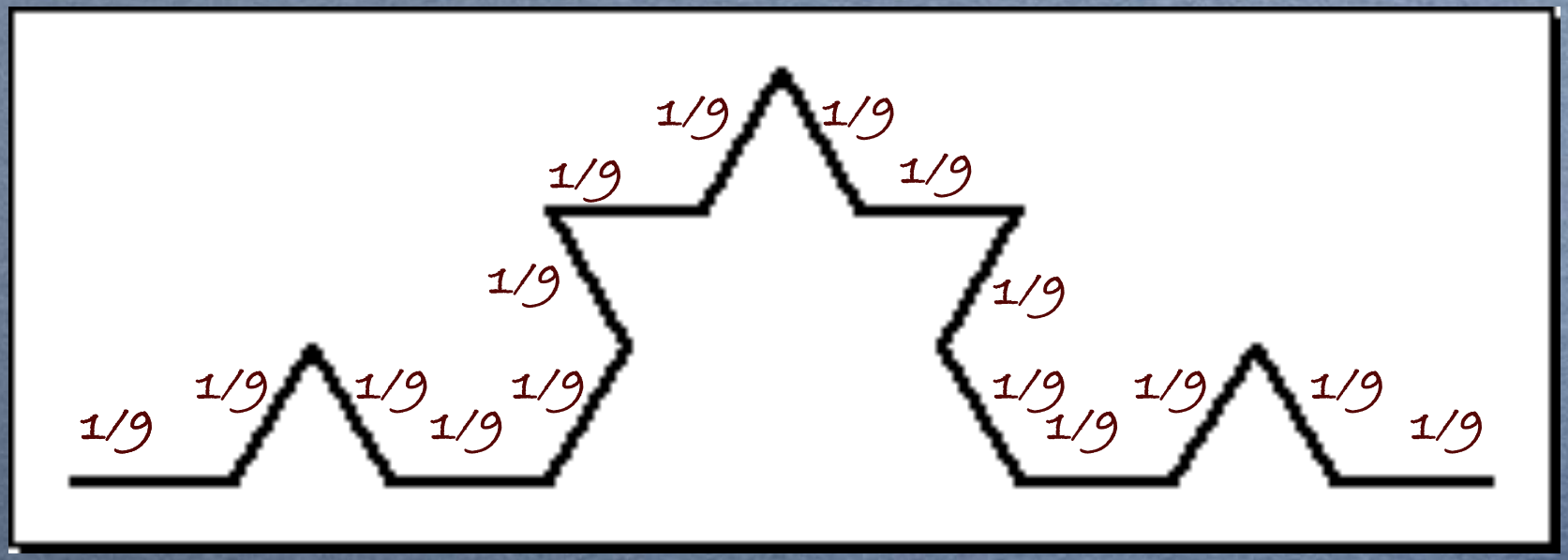
First step





Second step




1/9
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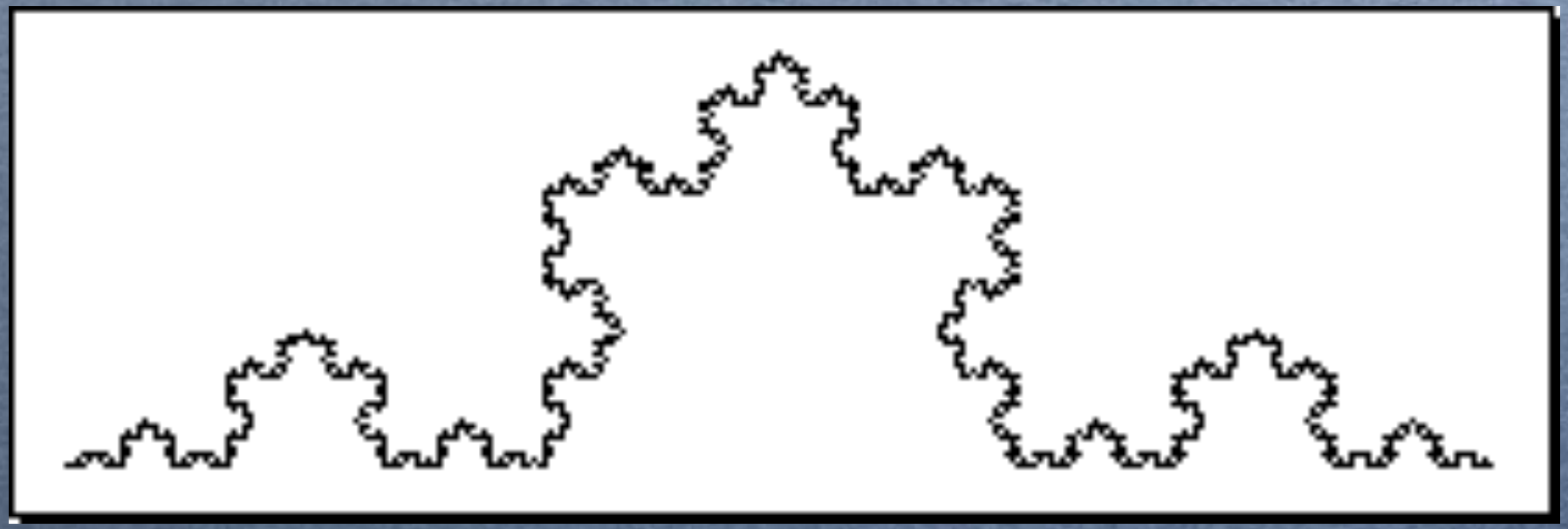
Third step

Each segment has size 1/27

....

The Koch Snowflake is the limit as $n \to \infty$

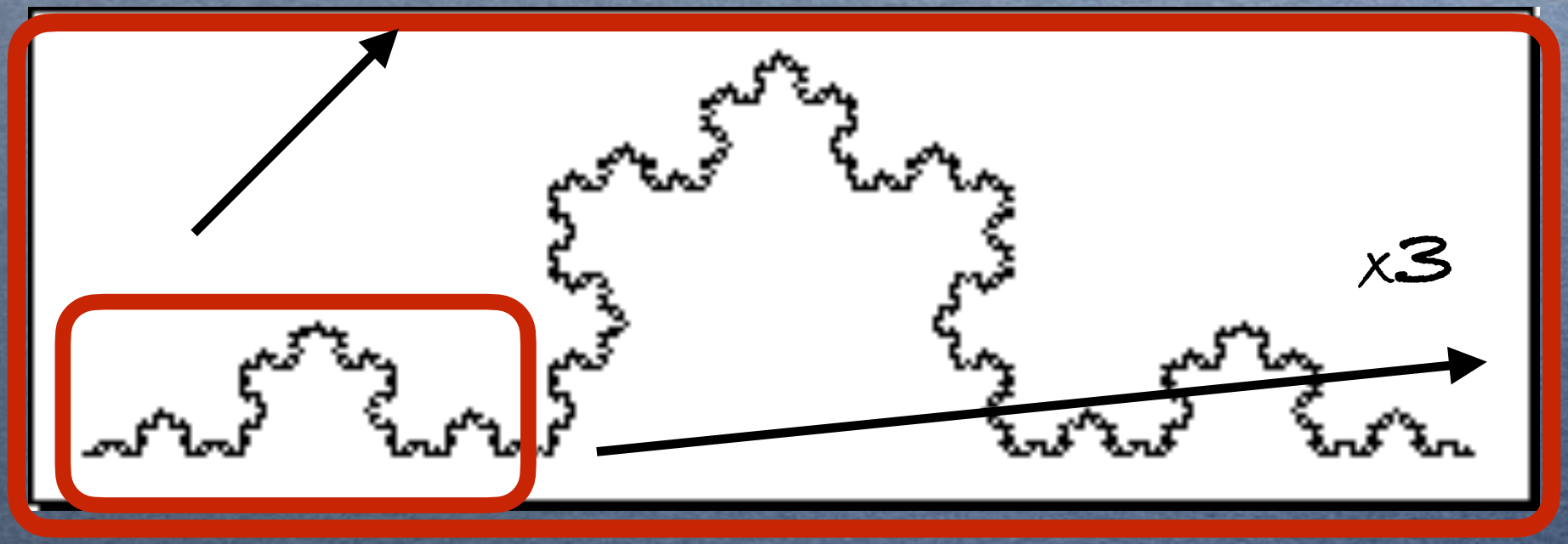
Let's try to calculate the dimension of the Koch Snowflake



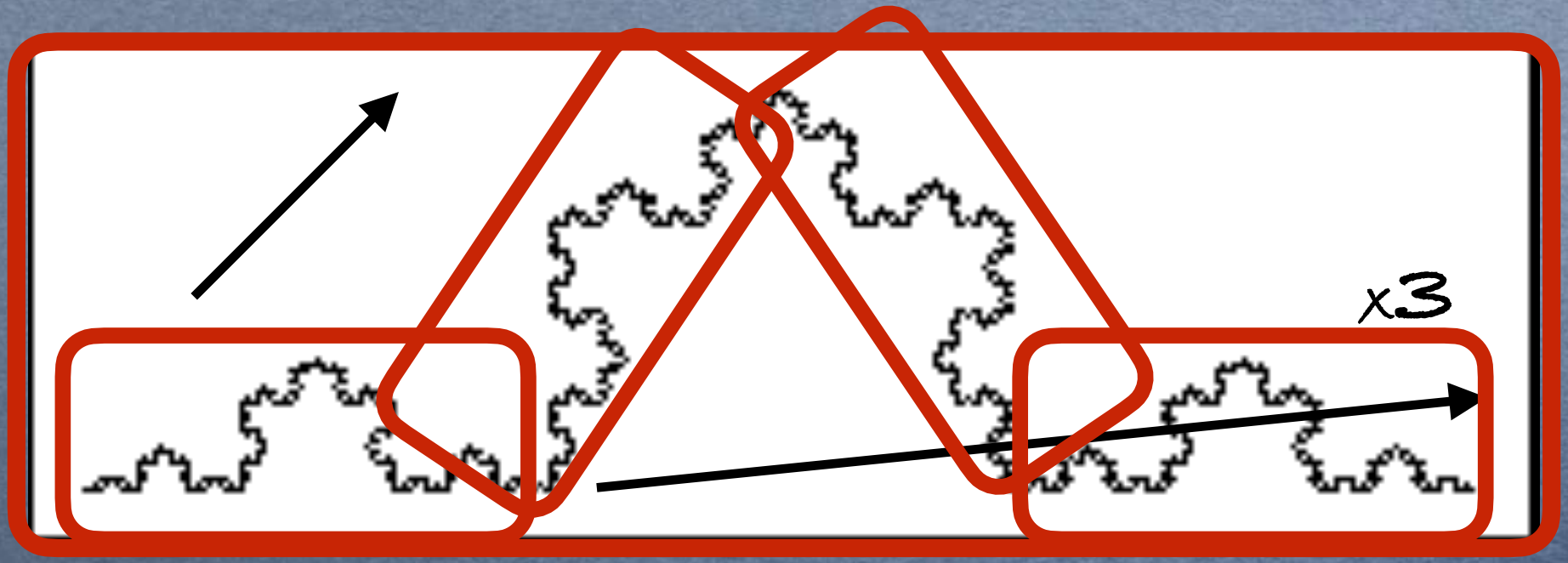
It looks like a "curve" of dimension 1

But we cannot assign any "coordinate" to any point because the distance between any two points is infinite!

Let's try with the new definition



The piece of curve in the small rectangle is an EXACT copy of the whole, but ONE-THIRD of the size

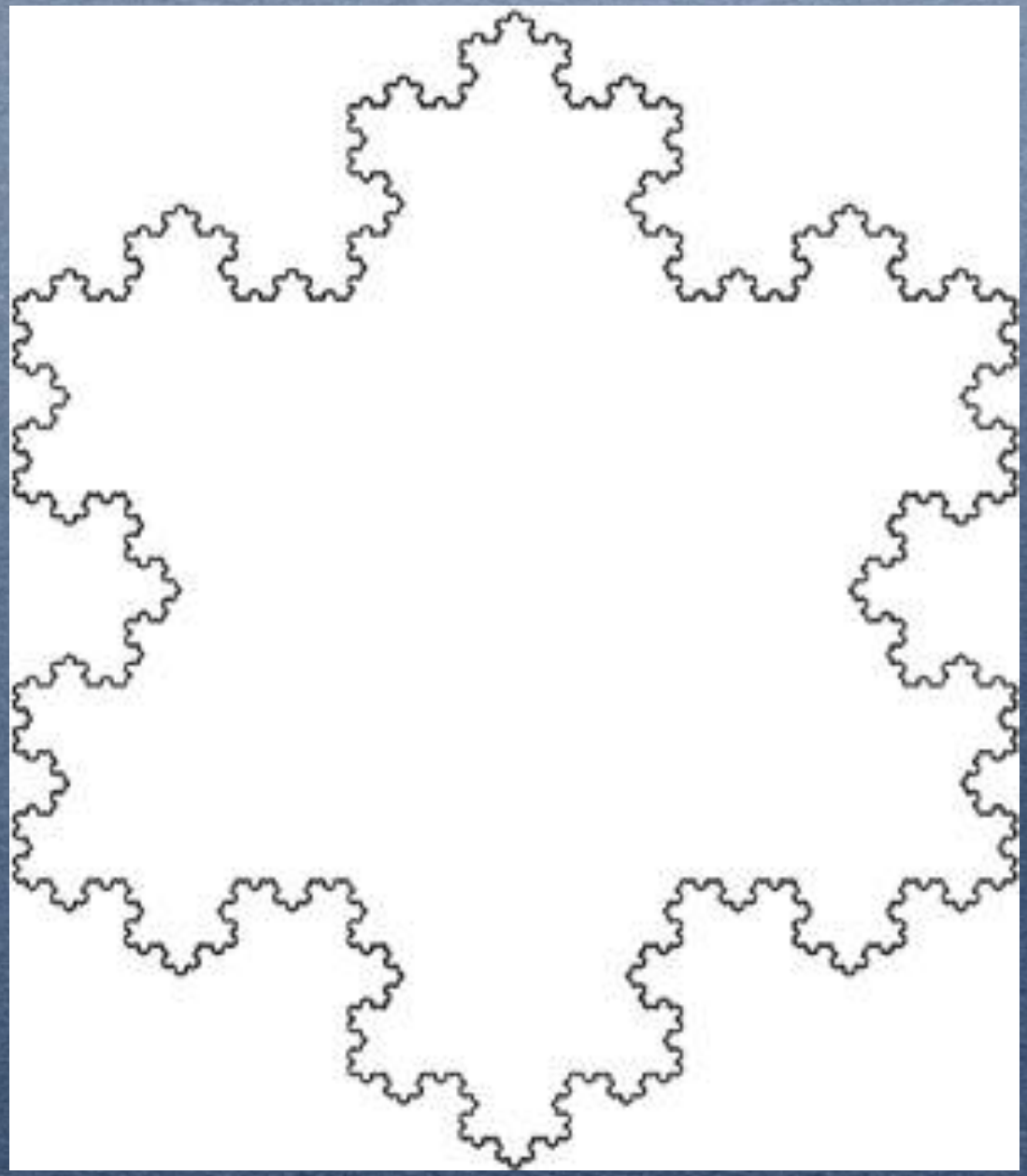


4 identical copies
scaled by a factor 3

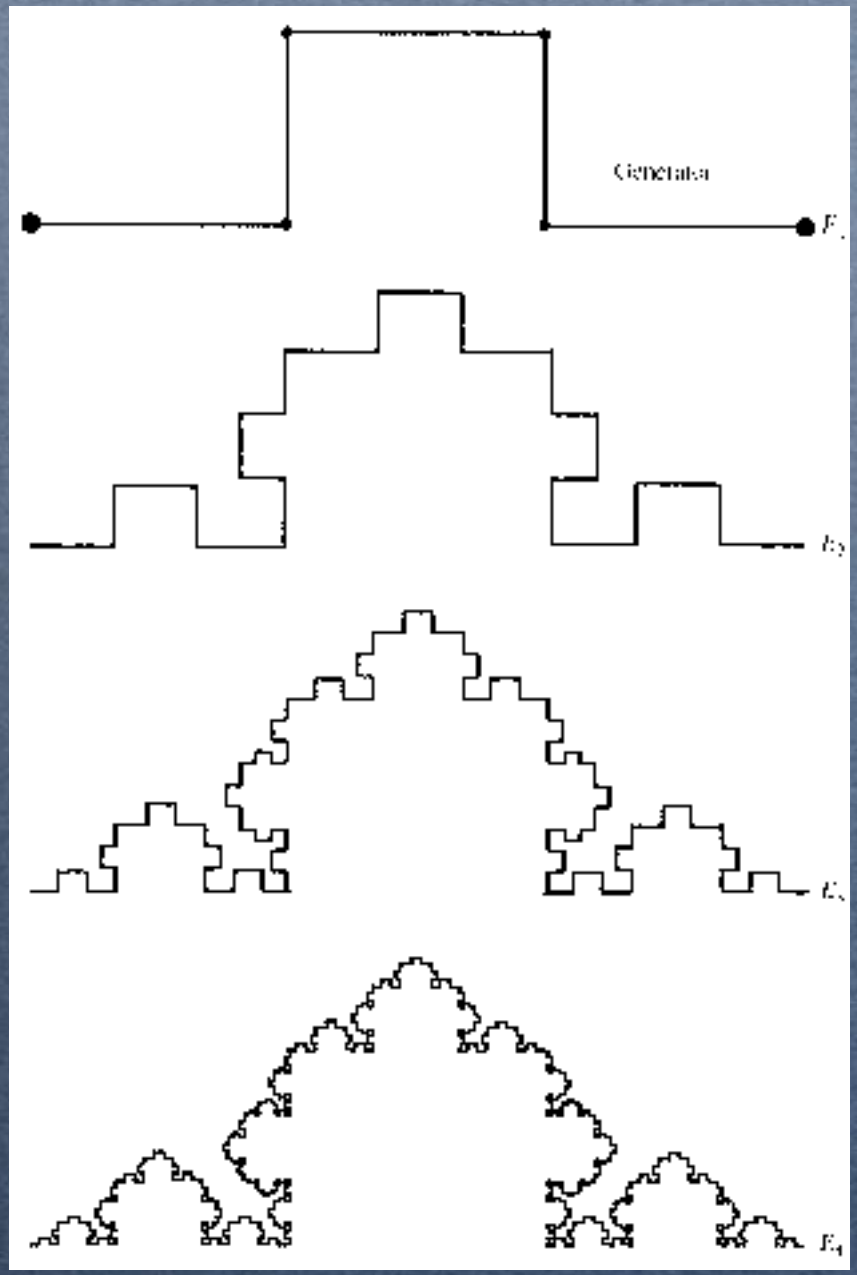
$$d = \log N / \log k = \log 4 / \log 3 = 1{,}261\ldots$$

The Koch Snowflake has dimension

$$d = \log 4/\log 3$$
$$= 1{,}261\ldots$$

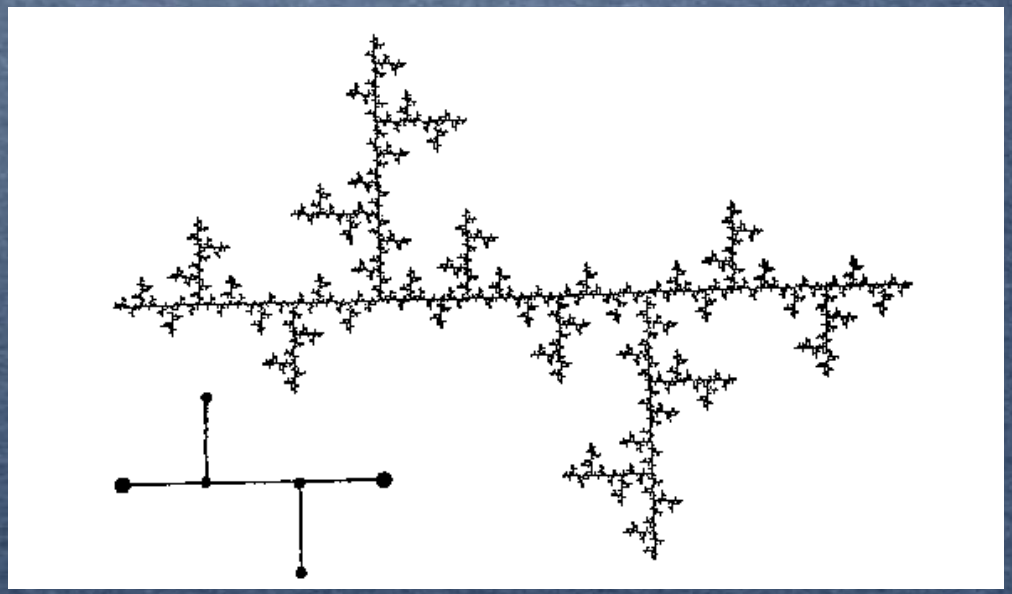


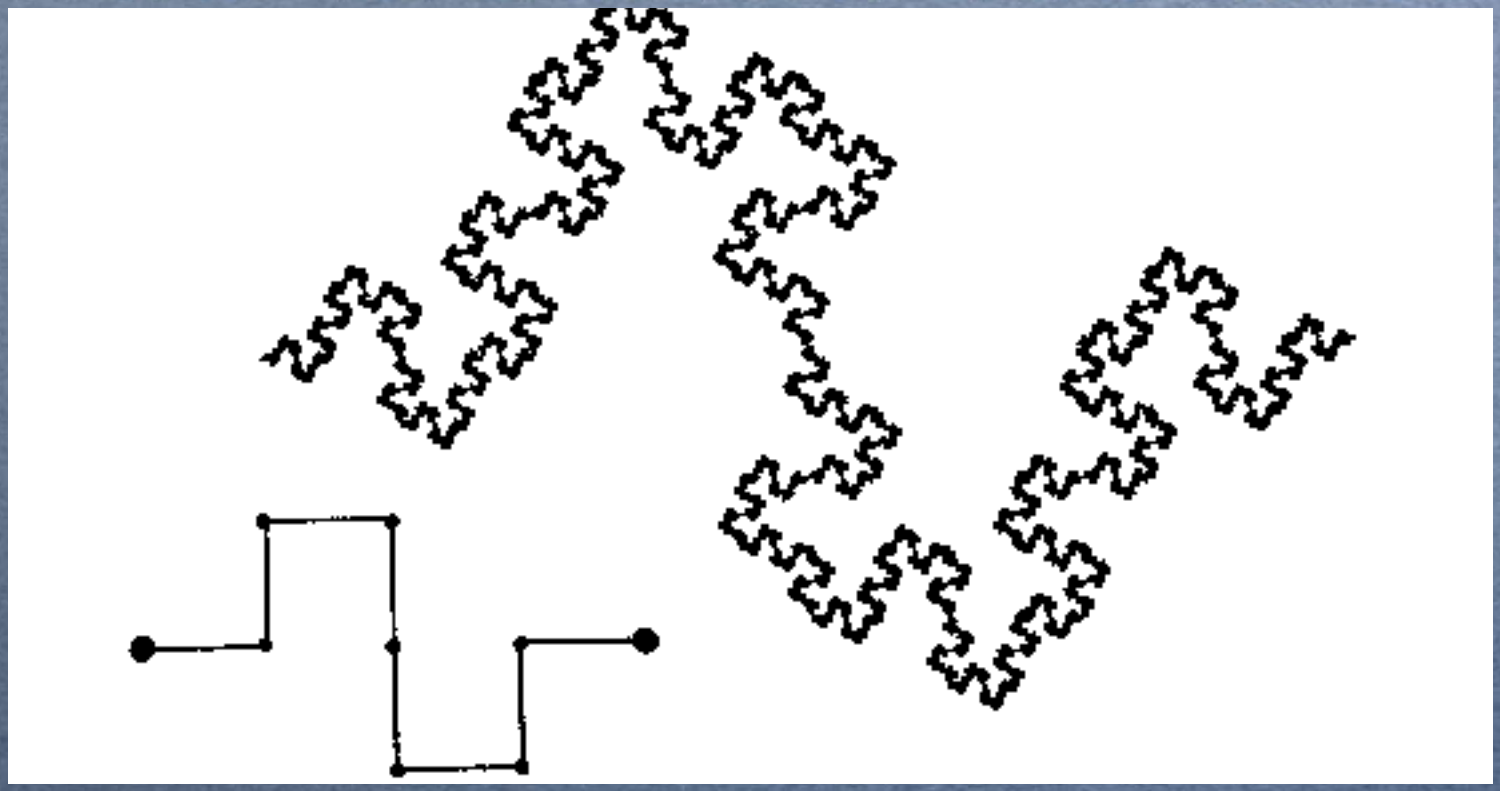
More examples



$$d = \log 5 / \log 3$$

$$= 1{,}464\ldots$$





$$d = \log 8 / \log 4 = \log 2^3 / \log 2^2$$

$$= 3 \log 2 / 2 \log 2$$

$$= 3/2 = 1{,}5$$

SELF-SIMILARITY IN NATURE

In the "real (physical) world" you cannot have a perfectly self-similar shape, just like you cannot have a perfect triangle or a perfect sphere.

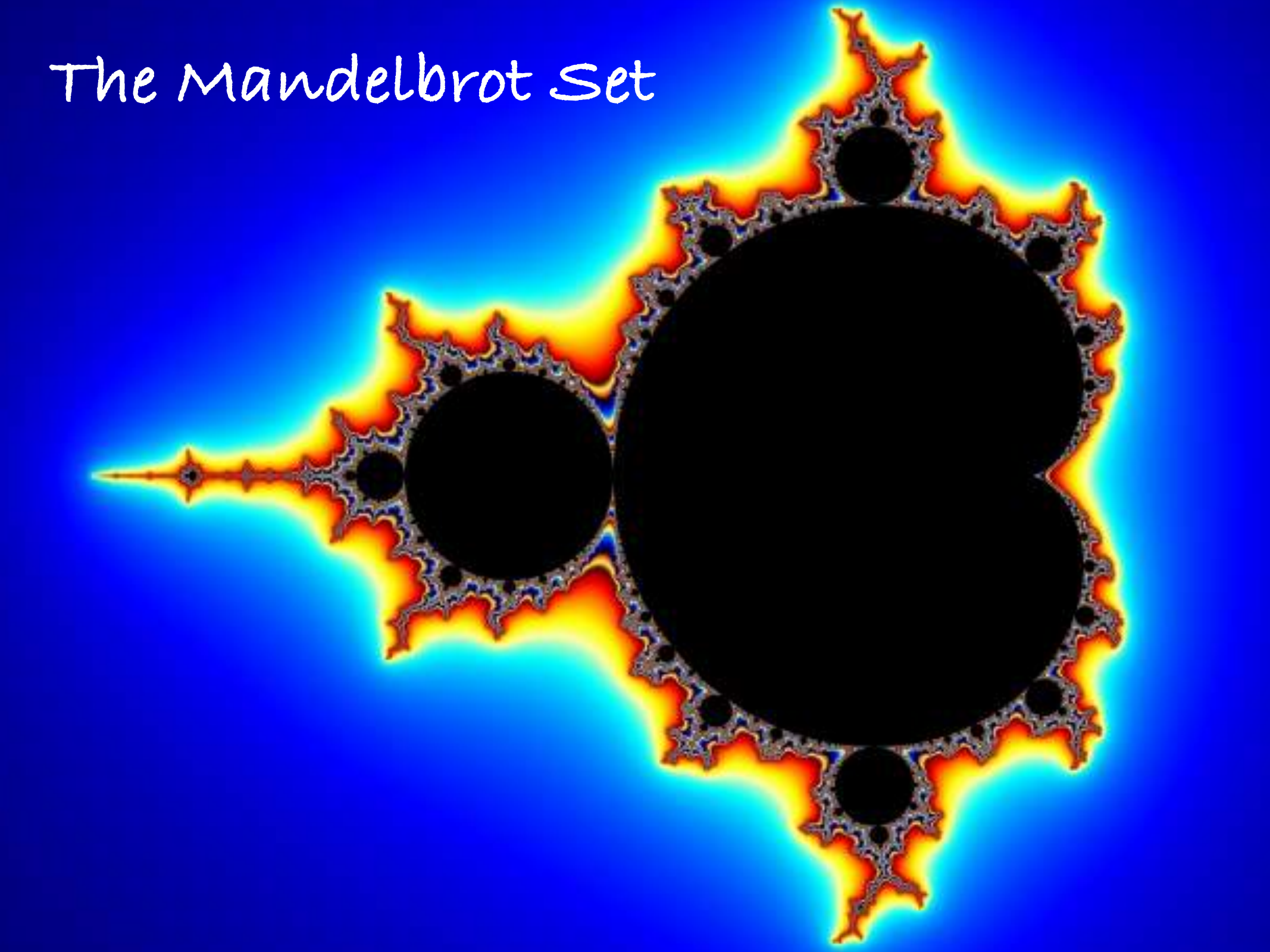
However, many objects in nature do exhibit some degree of self-similarity

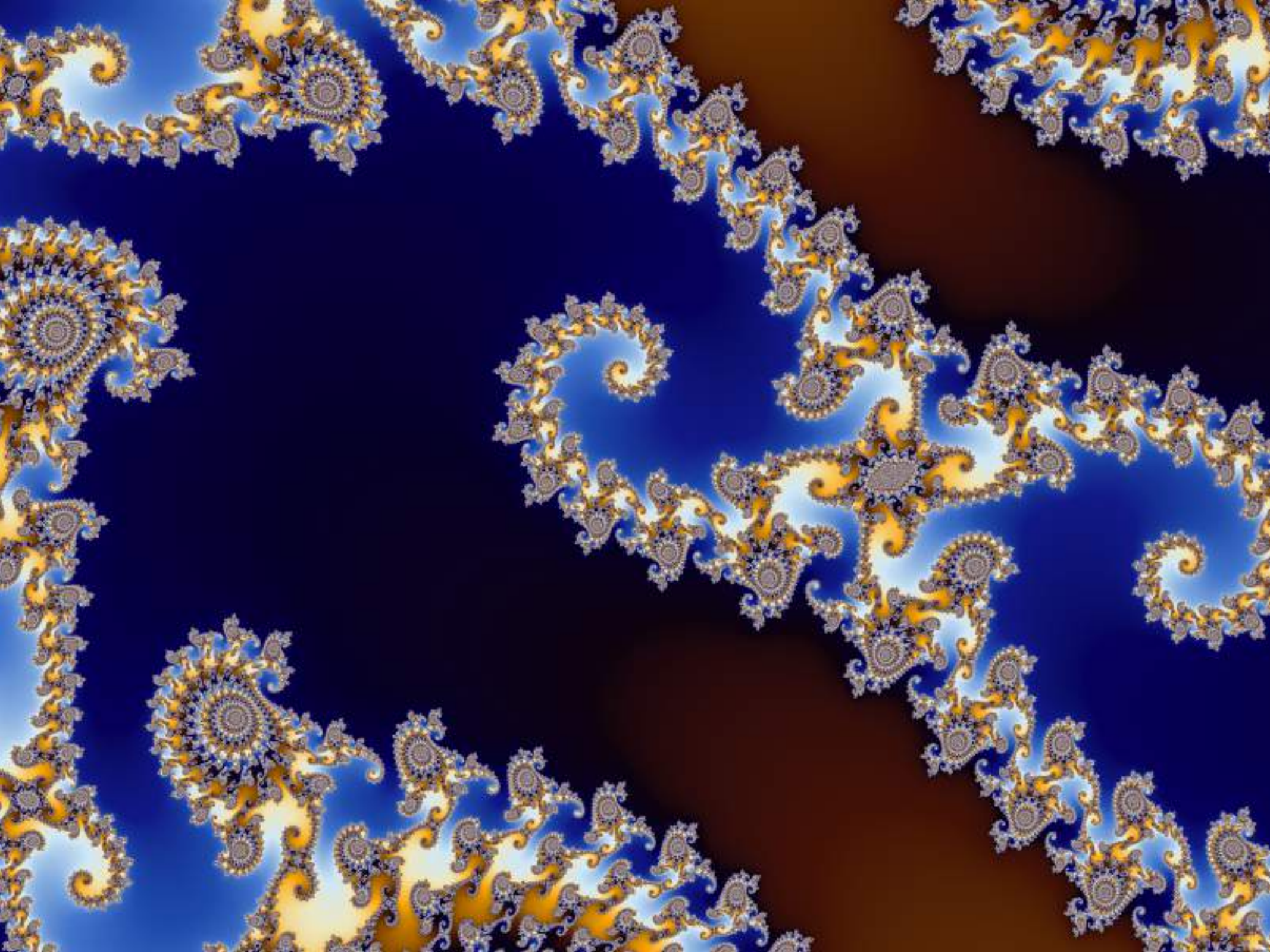
NON SELF-SIMILAR FRACTALS

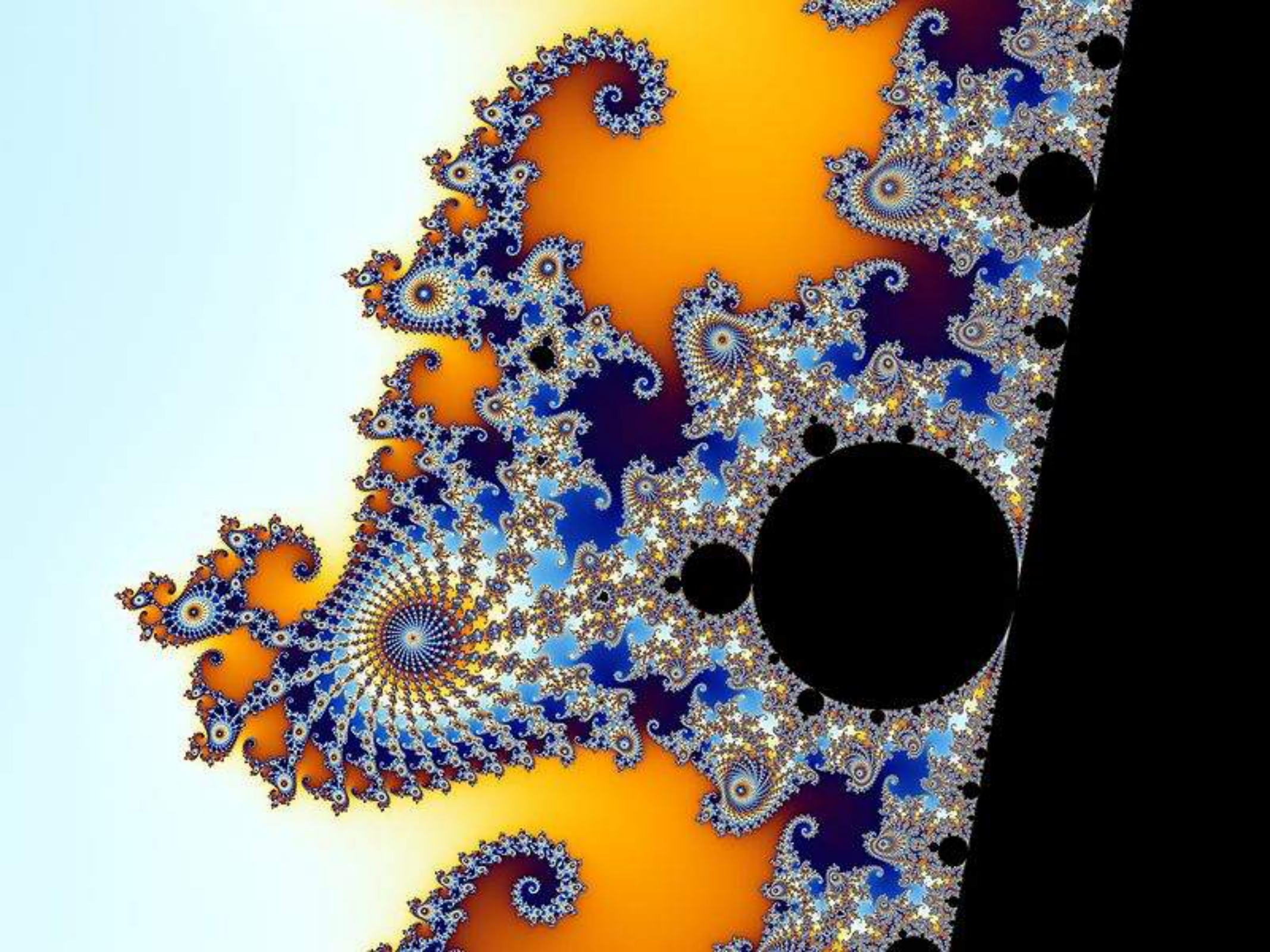
The mathematical examples above are "perfectly" self-similar in that they have infinite detail and have the SAME SHAPE AT EVERY SCALE

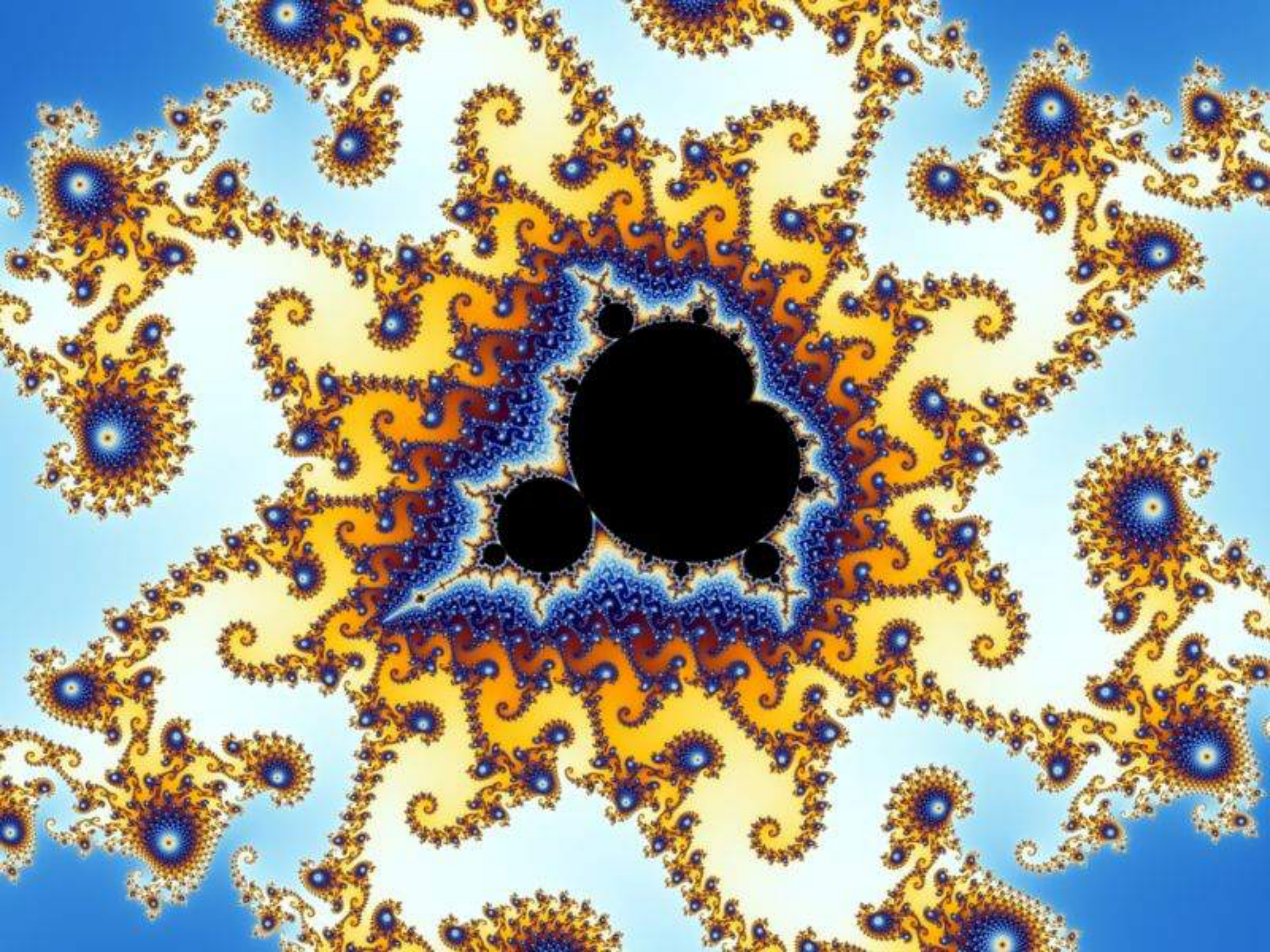
It is possible to construct examples which have infinite detail but have DIFFERENT SHAPES AT DIFFERENT SCALES

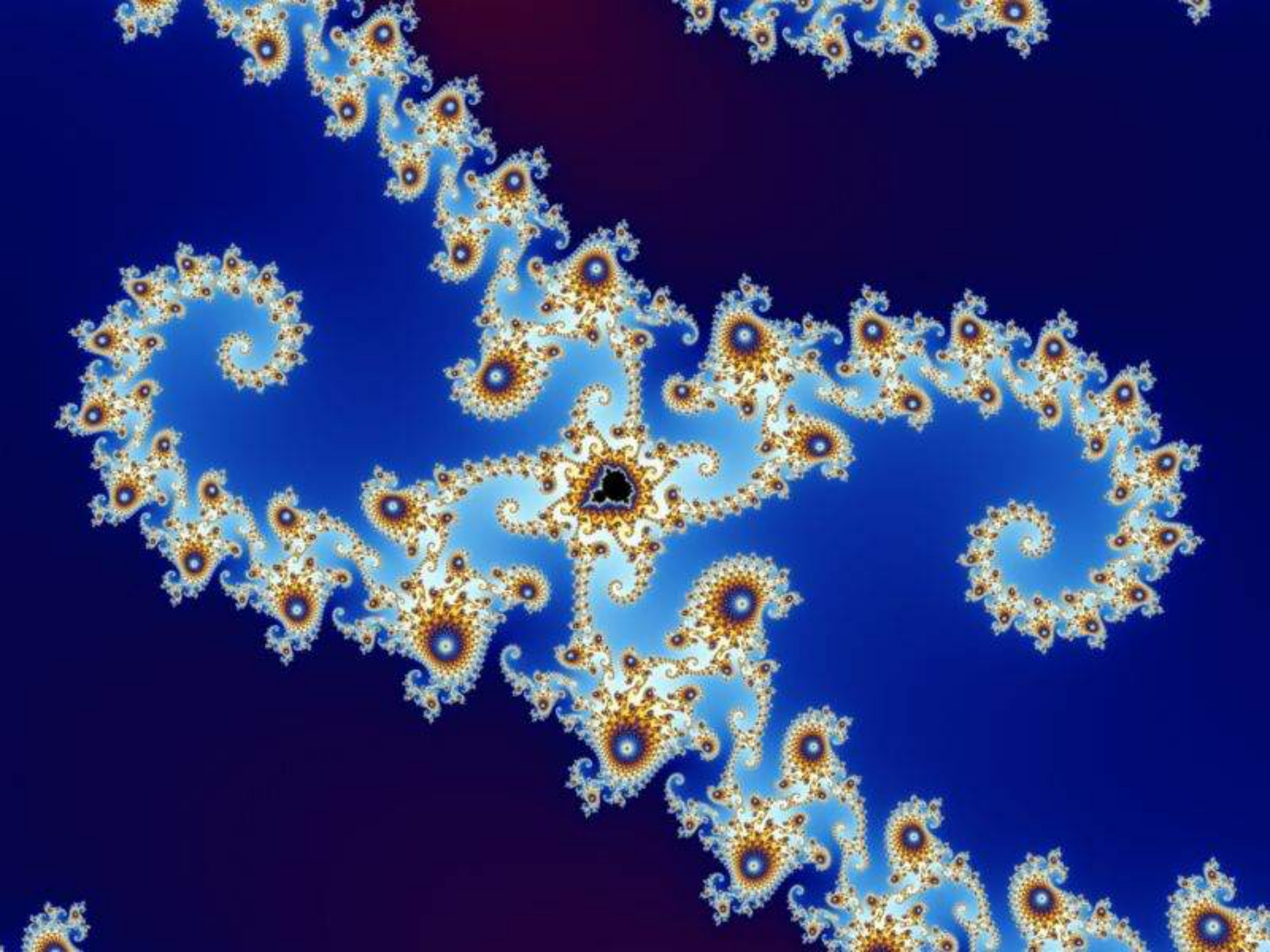
The Mandelbrot Set

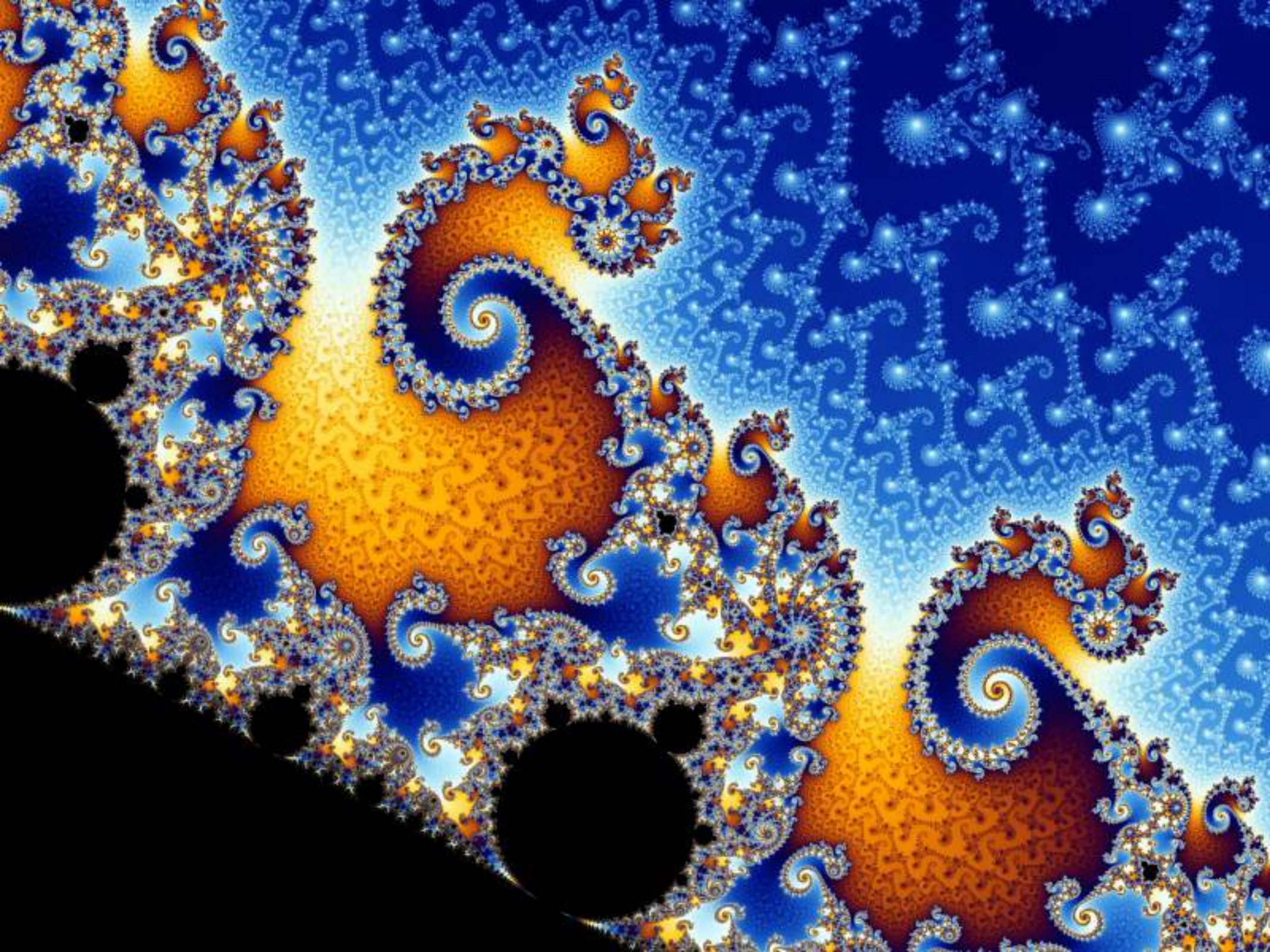


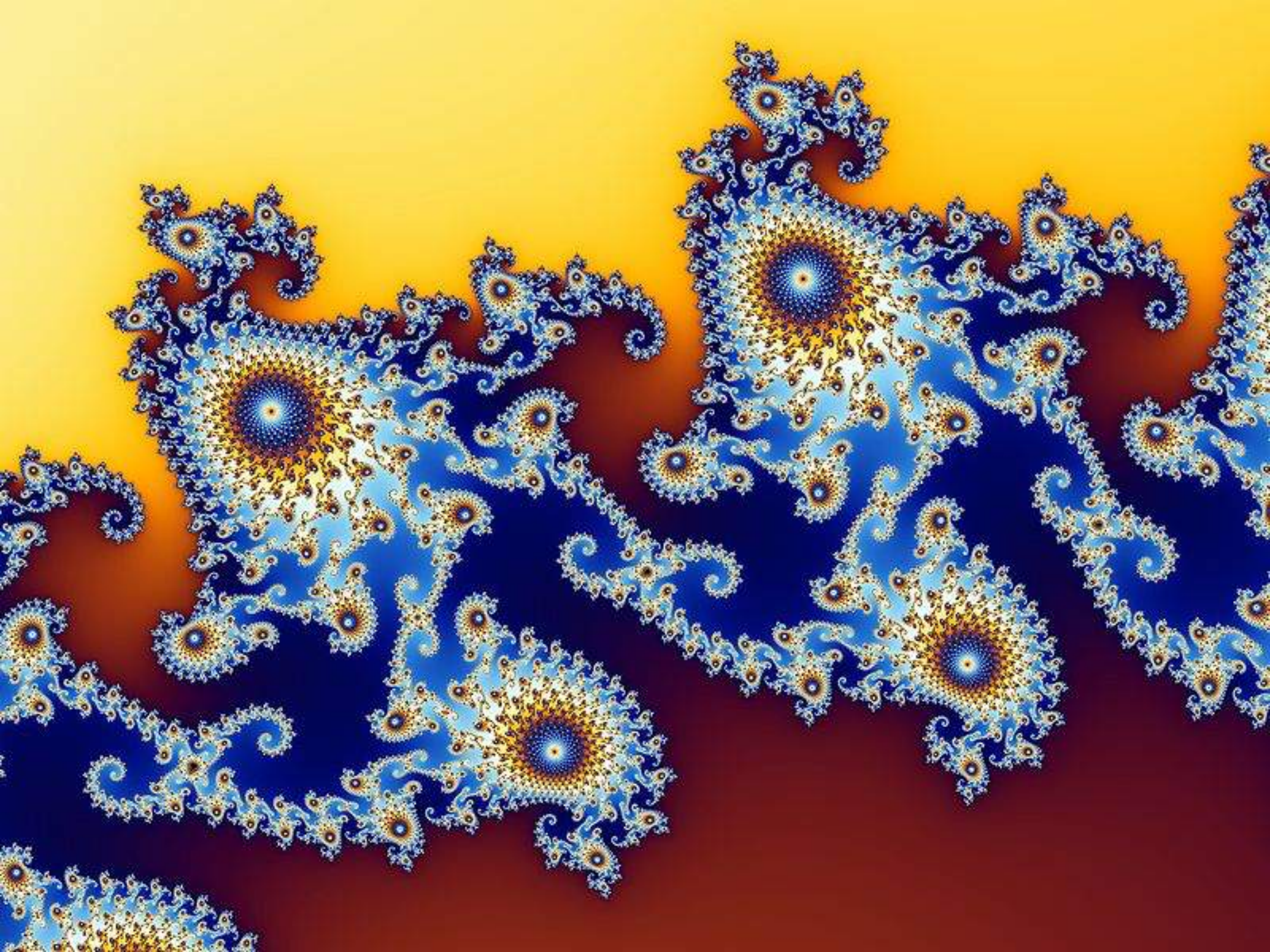


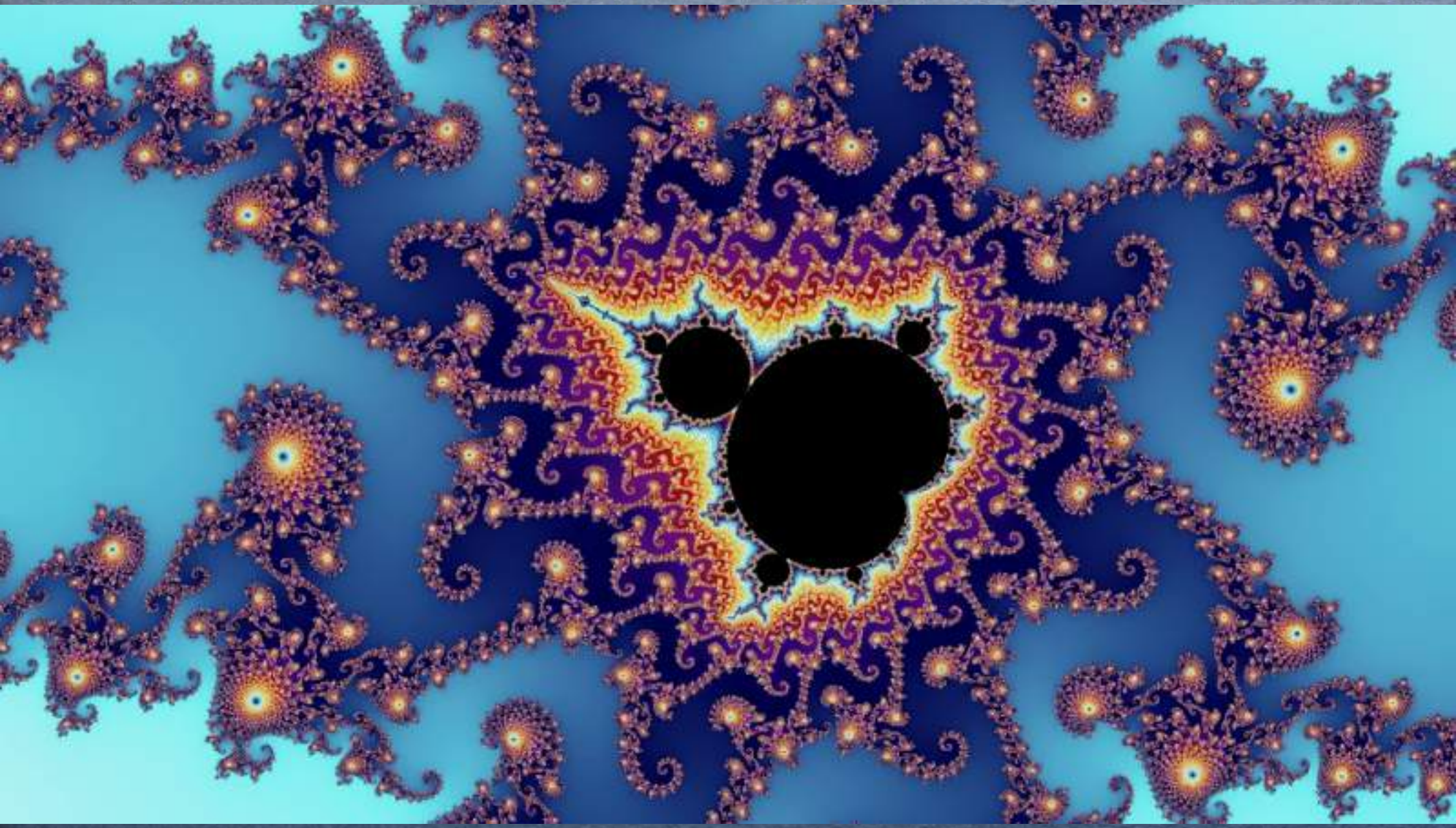












Thank you!

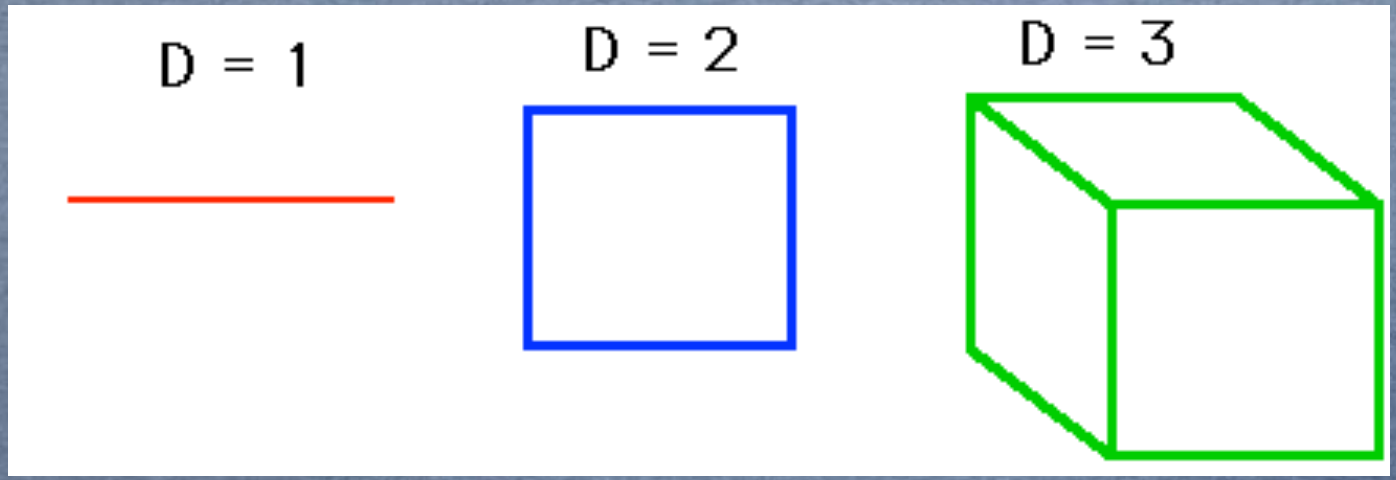
Hausdorff Dimension (1918).

Definition applies to any set

Involves the notion of "measure"

Felix Hausdorff (1868-1942)

How do you measure "size" ?



Segment → Length

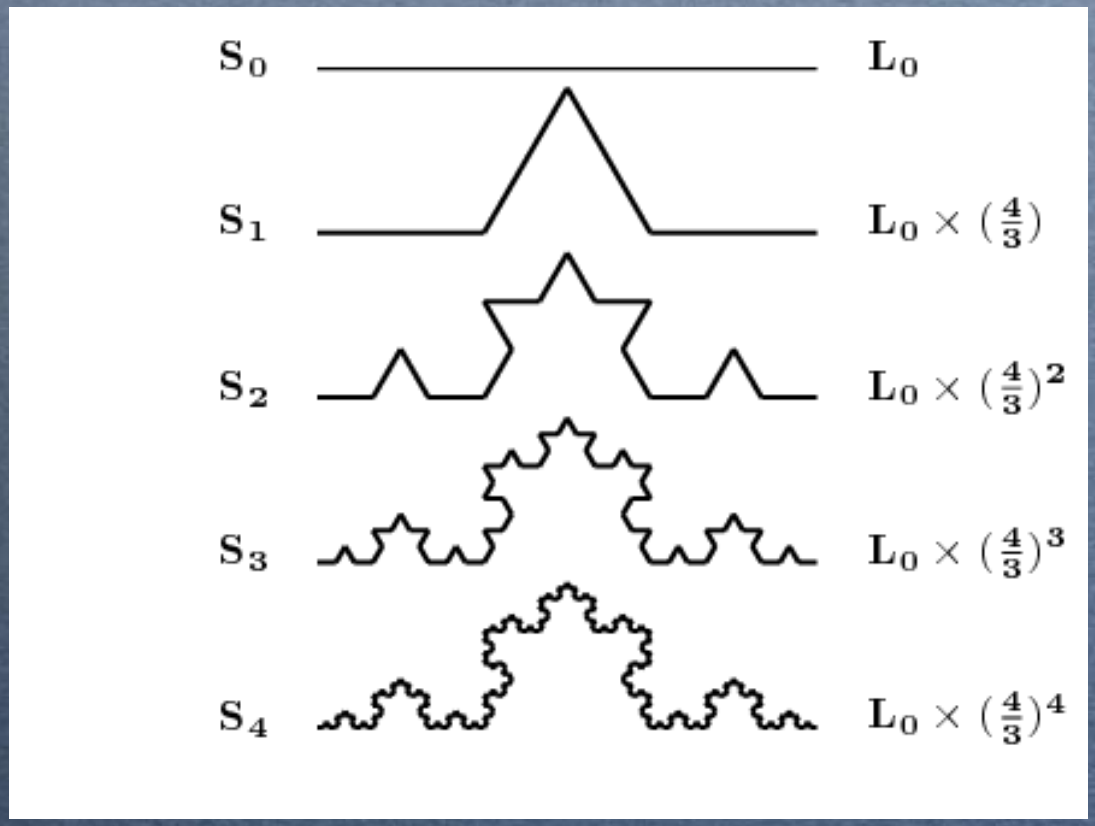
Square → area

Cube → volume

Each dimension has it own "measure"

A segment has zero "area", a square has infinite "length" but zero "volume"

What's the "size" of the Koch snowflake?



The length becomes larger and larger at each step, the final length is infinite

Need to measure it in a way that is appropriate for its dimension d=1,261

Hausdorff defined a general formula for the s-dimensional (Hausdorff) <u>measure</u> of a set F

$$\mathcal{H}^s(F) = \lim_{\delta \to 0} \left\{ \inf \left\{ \sum_{i=0}^{\infty} |U_i|^s : \{U_i\} \text{ is a } \delta\text{-cover of } F \right\} \right\}$$

It can be proved mathematically that, unless F is a finite set of points, there exist $s_F > 0$ such that:

$$H^s(F) = \infty \text{ if } s < s_F$$

$$H^s(F) = 0 \text{ if } s > s_F$$

$$\boxed{s_F = \text{Hausdorff Dimension of } F}$$

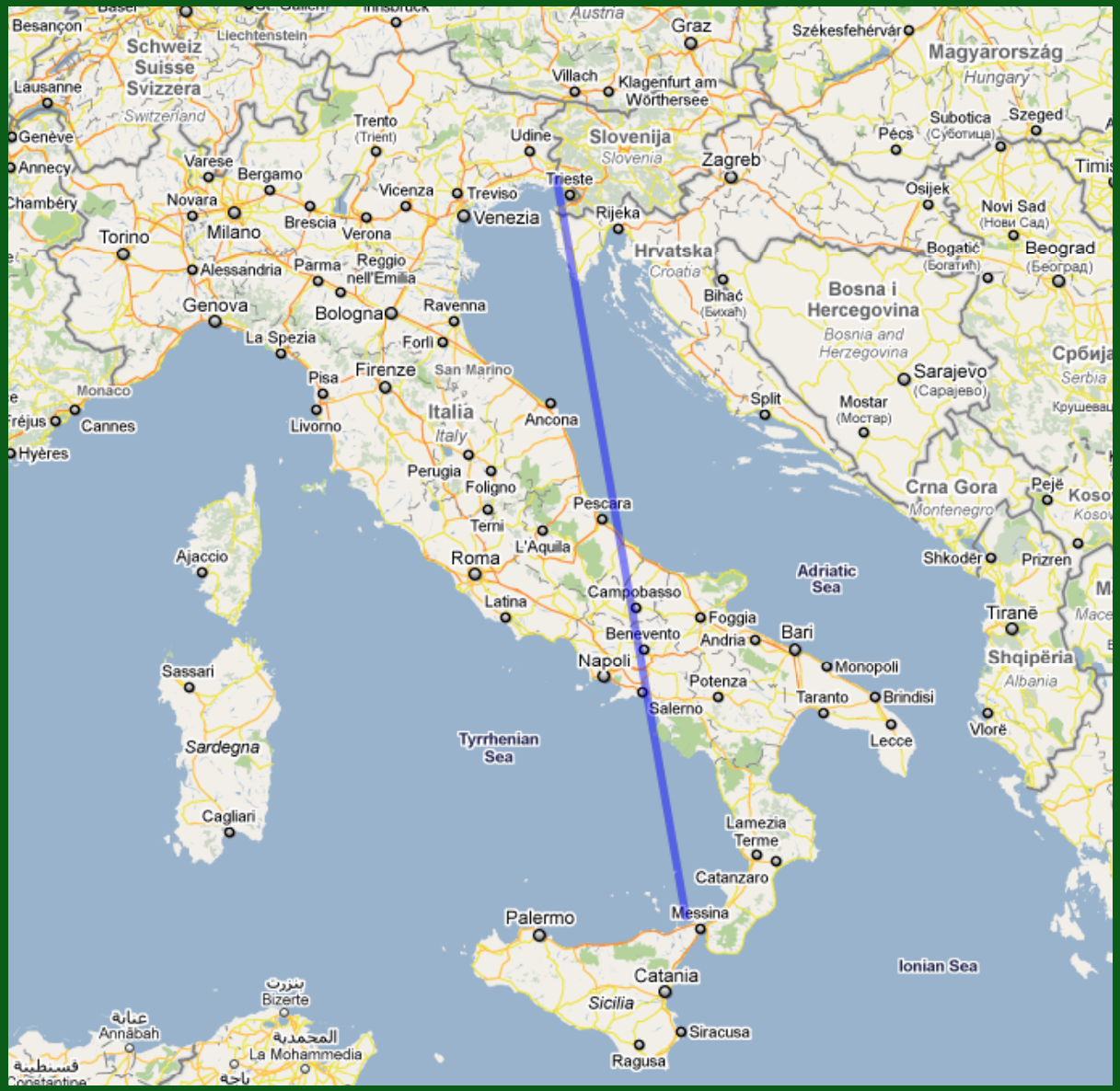
How long is the coast of Britain?

Using this more general notion of dimension one can try to estimate the dimension, and corresponding "measure", of objects in nature.

B. Mandelbrot
Science, vol 156,
pag. 636-638 (1967)

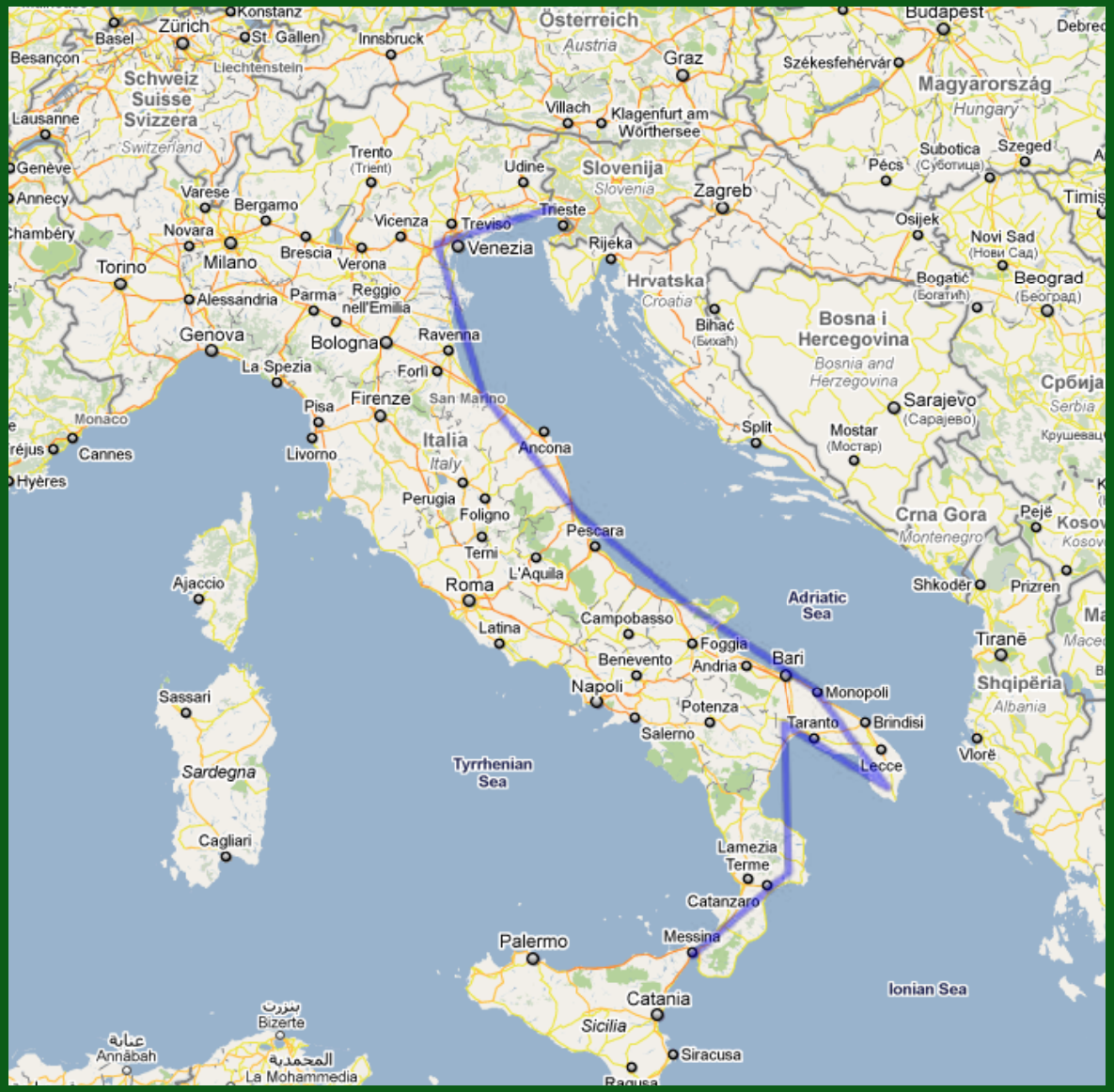
How long is the coast of Italy between Trieste and Messina?

The coastline paradox



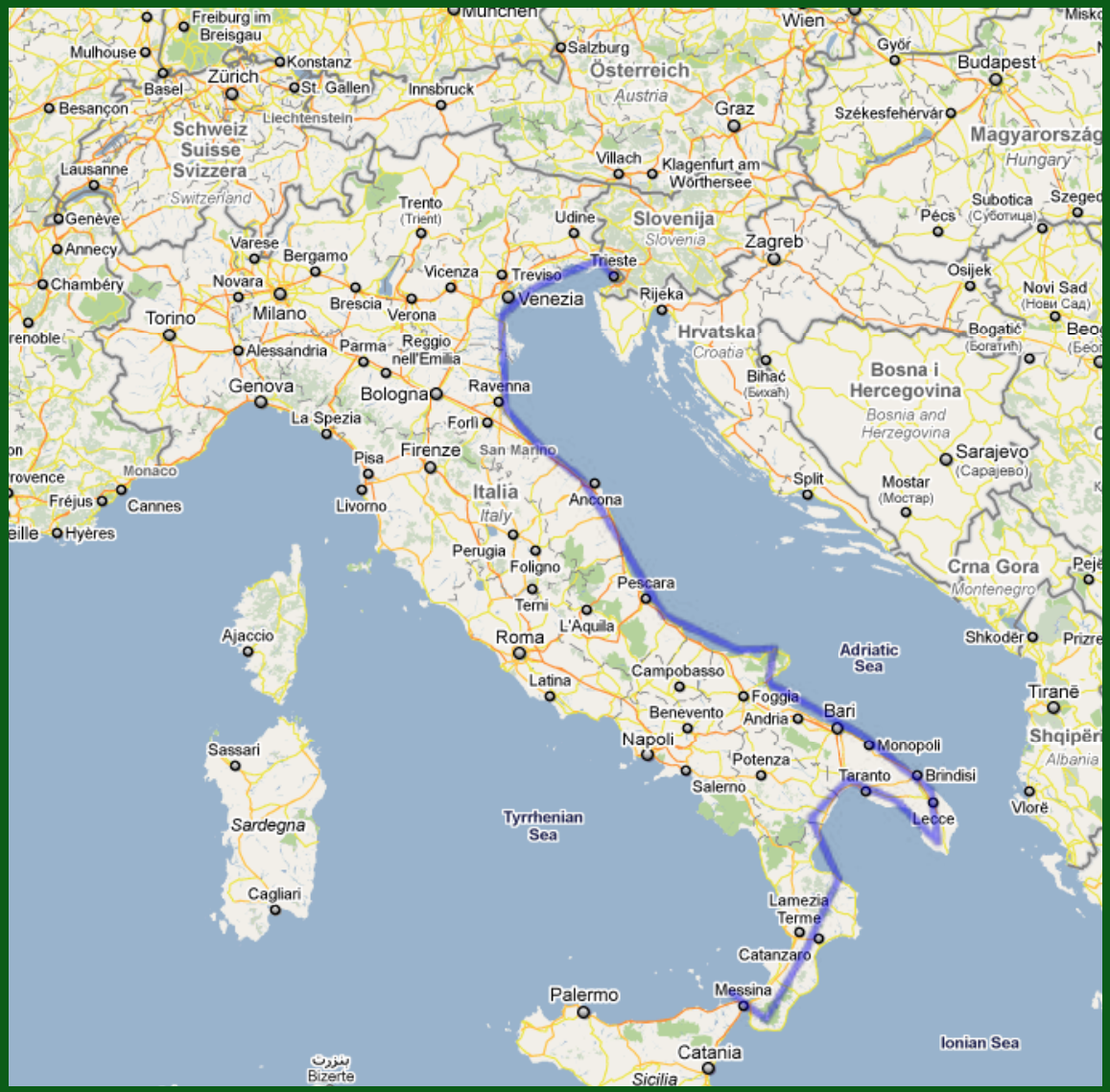
Trieste-Messina

First Approximation: 843 km



Trieste-Messina

Second Approximation: 1416 km



Trieste-Messina

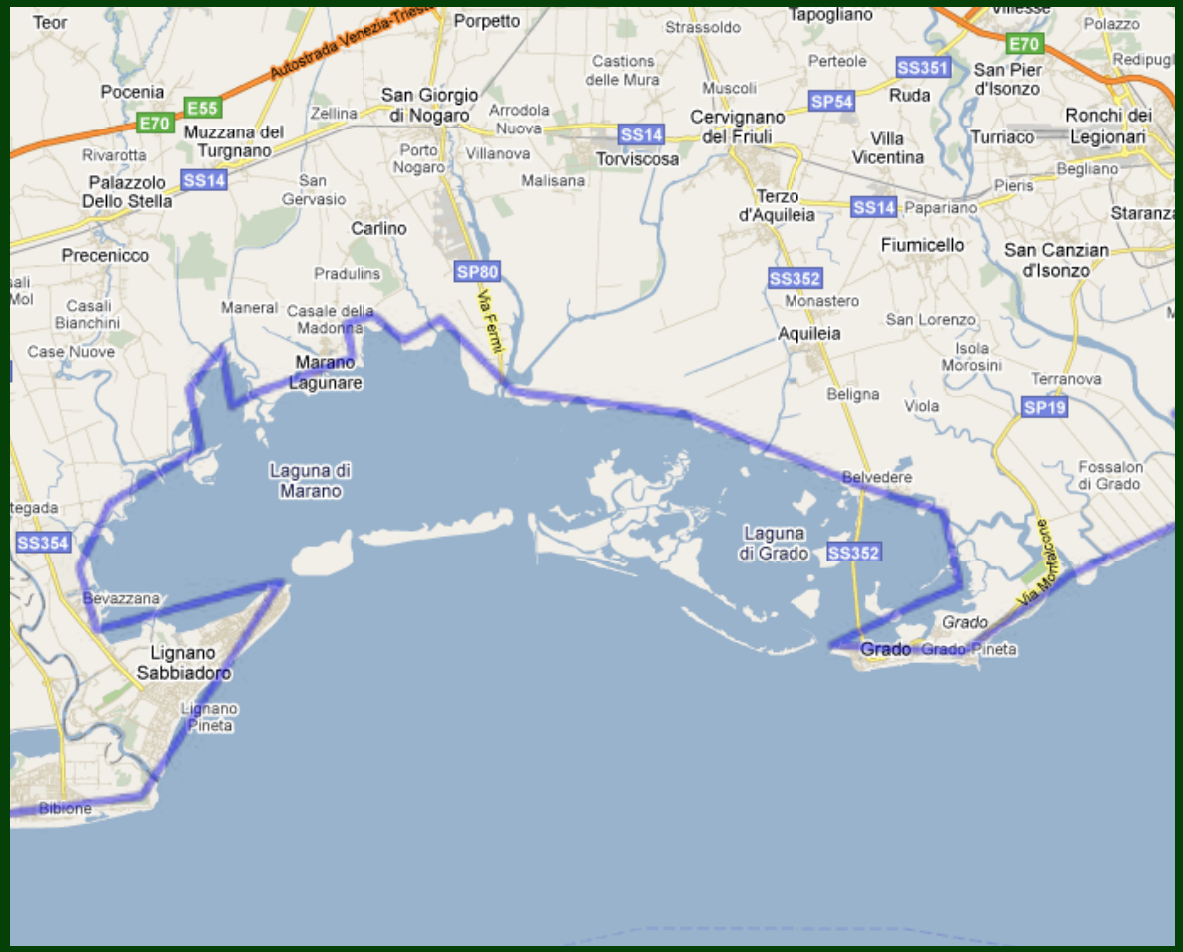
Third Approximation: 1506 km

Trieste-Messina

Fourth Approximation: 1696 km

Trieste-Messina

Fourth Approximation: detail



Trieste-Messina

Fifth Approximation:

...

How long is the coast?

At small scale it is not even clear where the land ends and the sea begins

So the length of the coastline depends on the scale at which it is measured

Mandelbrot's Suggestion was to

1) think of the coastline as a fractal
2) Calculate (approximate) dimension
3) Use Hausdorff measure

According to Wikipedia:

Dimension of Coast of Britain = 1.43

Hausdorff measure = 28,000 km

Constructing Fractal Sets

Fractal Objects
occur very naturally in Mathematics,
in particular in Dynamical Systems

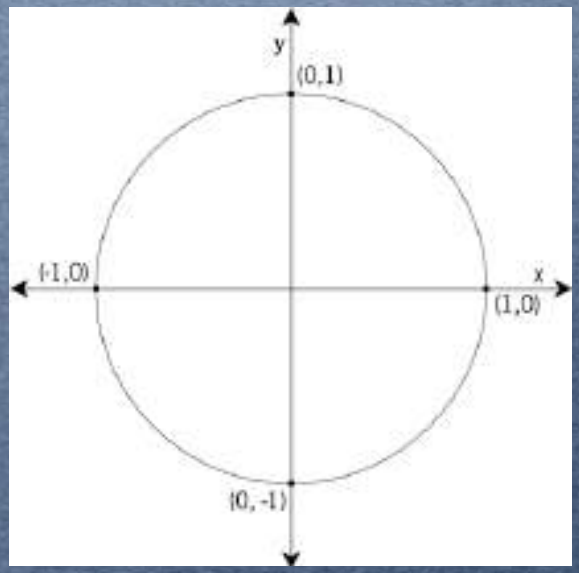
$\mathbb{C}$ = complex plane

$f: \mathbb{C} \to \mathbb{C} \quad f(z) = z^2$

If $|z| > 1$ then $|f(z)| > |z|$

If $|z| < 1$ then $|f(z)| < |z|$

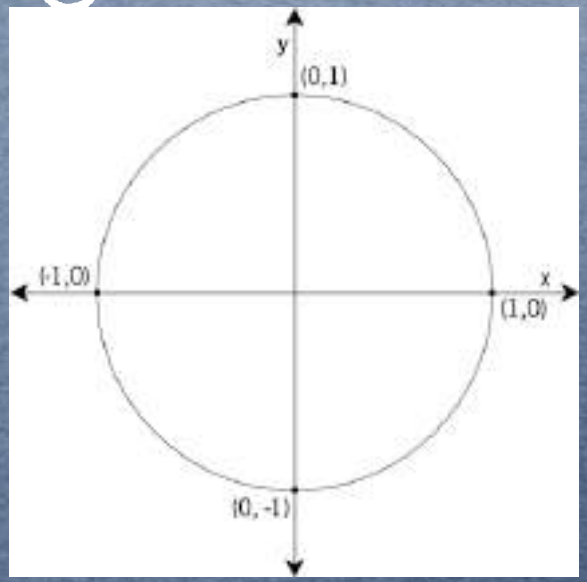
If $|z| = 1$ then $|f(z)| = |z|$



$J(f)$ = Julia set of f = S^1 = topological boundary of set of points whose future orbits remain bounded

Now consider c in $\mathbb{C}$ very small

$f_c(z) = z^2 + c$



The orbits of points far from the unit circle still behave as before

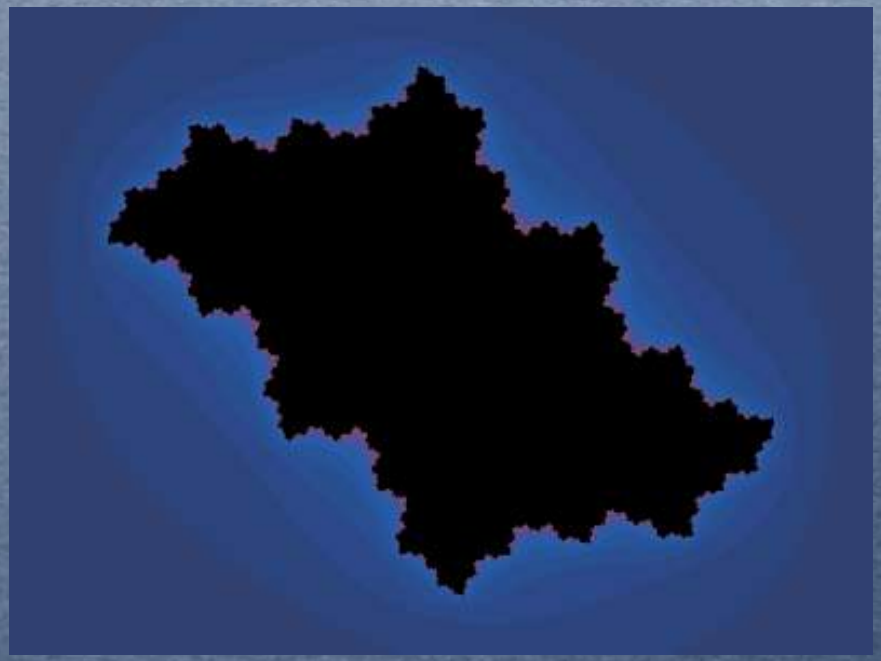
Points near the unit circle are more unpredictable

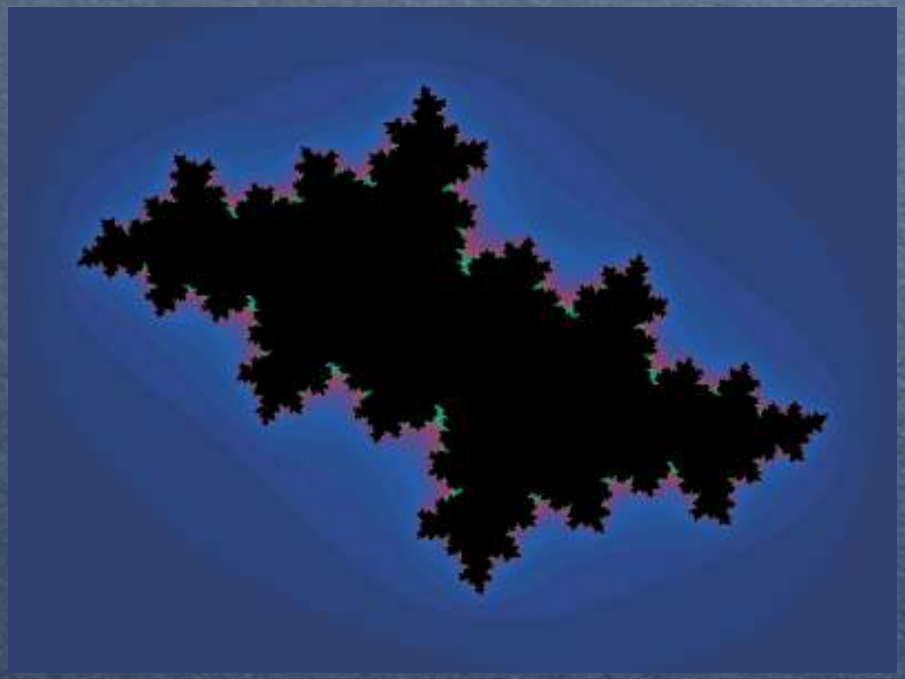
Julia set $J(f_c)$ is a closed "curve" with dimension >1

$f_c(z) = z^2 + c$

Julia sets

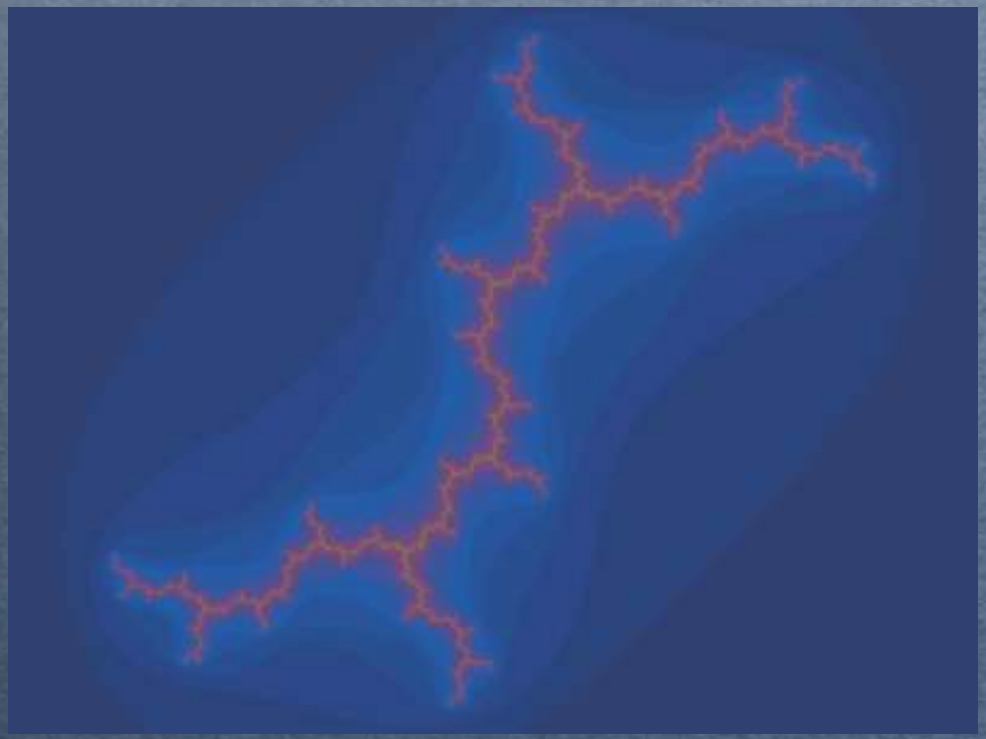
$c = 0.25 + 0.52i$



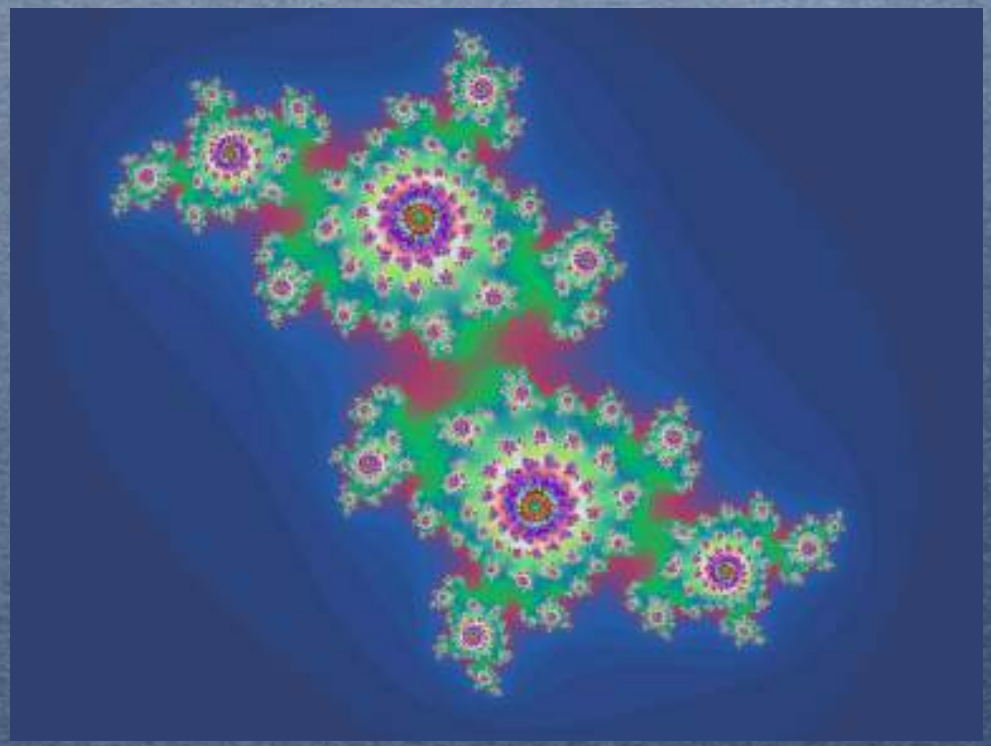


$c = -0.5 + 0.5i$

Dendrite Julia set

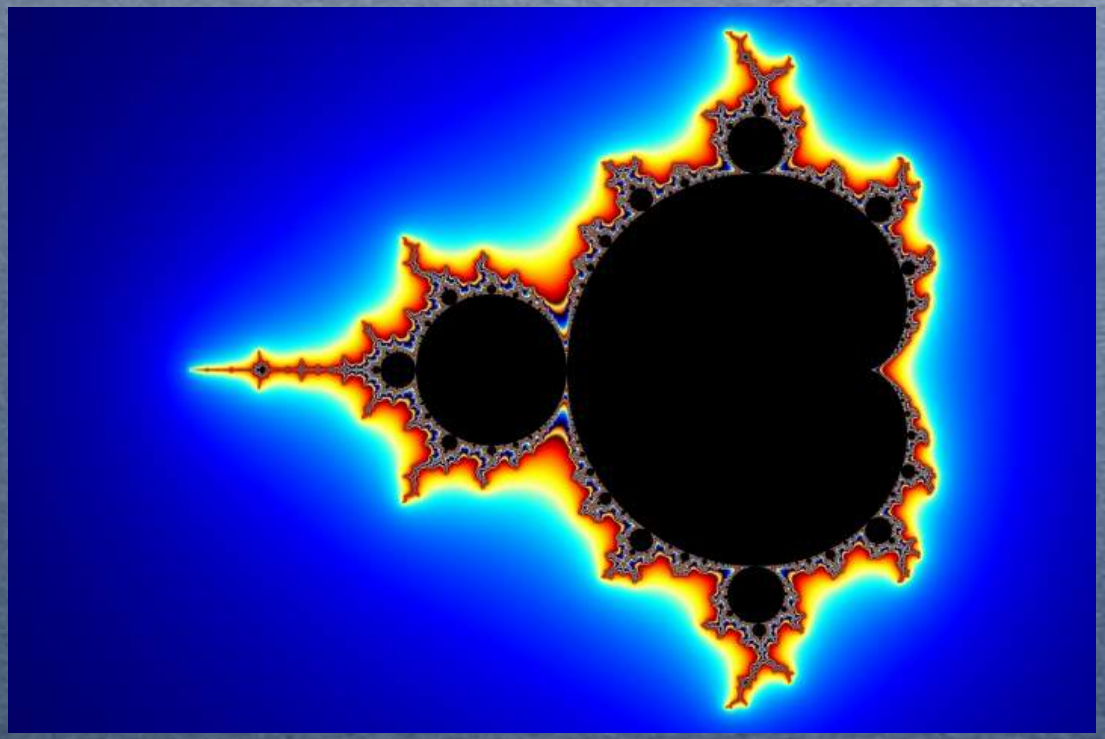


Cantor Julia set



Every Julia set is either connected or totally disconnected

Mandelbrot Set = set of parameters c
for which Julia set is connected



Shishikura (1998, Annals of Math)
Boundary of Mandelbrot set
has Hausdorff dimension 2